Financial Accelerator with Specialized Lenders *

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October 15, 2026

Preliminary. Comments are welcome.

Abstract

We develop a model of credit cycles in which specialized lenders can preserve borrower going-concern value after foreclosure, allowing firms to borrow against value that arm's-length financiers cannot fully realize. Limited pledgeability arises because this recovery advantage cannot be fully transferred to the intermediary's own creditors, so specialized lending requires intermediary balance-sheet resources. With heterogeneous borrowers, specialized intermediaries expand credit especially to firms with less transferable, more intangible value. Following sufficiently large adverse shocks, however, scarce intermediary resources induce a *flight-to-pledgeability*, shifting new lending toward borrowers with higher recovery values. We then apply the model to private equity and private credit markets and show that the presence of specialized intermediaries can explain several documented facts about lending cycles and private credit specialization. We also test and find supportive evidence for flight-to-pledgeability, using Business Development Company lending during COVID-19 as our laboratory.

Keywords: Private Credit, Financial Accelerator, Business Development Companies (BDCs)
JEL classification: G21, G32, G34

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1 Introduction

Private credit has expanded rapidly as an alternative to bank lending and public debt markets. Yet private-credit loans generally carry higher interest rates than bank loans, syndicated leveraged loans, and public corporate bonds.[1] Why do borrowers nevertheless choose such specialized lenders when other forms of bank or market-based financing are available? One answer is that specialized lenders can lend against forms of value that are difficult to transfer. A specialized lender may know how to restructure a distressed borrower, operate collateral, preserve customer relationships, or maintain a productive asset after default. These capabilities can make collateral worth more to such a lender than to an outside investor, allowing the lender to extend more credit against the same underlying assets.

This paper studies the macro-financial consequences of that recovery advantage. Our starting point is a distinction between two forms of collateral value. Some value is transferable: it can be realized by essentially any creditor through liquidation or resale. Other value can be realized only by a lender with the relevant expertise or operational capacity to keep the borrower's business operating. A specialized lender may therefore recover substantially more from a distressed borrower than an outside financier. Crucially, however, the specialized lender cannot pledge the entire recovery value to its own creditors. It can pledge only the transferable component[2], as outside lenders can recover only the market value of the assets in case of liquidation. Specialized lending therefore creates additional borrowing capacity for firms, but doing so requires the specialized intermediary to commit its own balance-sheet resources.

We introduce this feature in a lumpy-investment economy in the spirit of Kiyotaki and Moore (1997).[3] Keeping with their agrarian metaphor, we refer to borrowers as farmers and outside financiers as gatherers. We add a third group of *gardeners*, specialized intermediaries that can operate seized projects after foreclosure. Farmers occasionally receive opportunities to invest by planting new trees on their land. The resulting stock of trees is productive but is generally not transferable with the land: if the project is liquidated to a gatherer, the trees are

[1] Chernenko et al. (2022) and Haque et al. (forthcoming) show that the premium over bank loans persists after controlling for borrower and loan characteristics, while Cai and Haque (2024) documents higher average spreads relative to institutional Term Loan B loans. IMF (2024) documents higher borrowing costs in private credit relative to syndicated leveraged loans and public corporate bonds.

[2] This idea relates to inalienability of human capital by Hart and Moore (1994).

[3] In Kiyotaki and Moore (1997) lumpy means that the investment opportunities arise intermittently.

lost and only bare land remains. Because investment opportunities are lumpy, this productive stock cannot be immediately rebuilt after the transfer. An incumbent gardener that forecloses can instead keep the trees alive and preserve their continuation value. Farmers can therefore pledge the continuation value of the land with its trees intact to gardeners, whereas gardeners can pledge only the market value of bare land to gatherers. The difference between these values varies with equilibrium market conditions. The additional credit supported by the continuation value of trees is backed by limited gardener equity, which makes specialized lending dependent on both market conditions and the gardener's balance sheet.

As long as the gardener can absorb losses from its current cash flows, it can continue lending against this non-transferable value. The resulting decline in borrower debt capacity is therefore smaller than in the standard Kiyotaki-Moore economy. The credit crunch can nonetheless be more persistent because borrowers remain more leveraged in this economy. With more debt to repay, a shock to borrower wealth leaves fewer internal resources available when subsequent investment opportunities arrive. Once the cash-flow buffer is exhausted and gardener equity becomes scarce,[4] the gardener extends less credit against the non-transferable component of recovery value, which it cannot pledge to gatherers. The effects of this non-transferable value therefore depend on the availability of gardener resources. It cushions borrowing after moderate shocks, but once those resources become scarce, borrowing contracts most for farmers that rely on it heavily. Lending consequently shifts toward farmers with more pledgeable collateral. We call this reallocation a flight to pledgeability.

Our model first clarifies when specialization matters. Suppose a seized project could change hands without loss. Its competitive market price already incorporates the value generated by its most productive use. An outside financier holding it could then sell it at the market price, and so would recover exactly what the specialist would have recovered by operating it; borrowers would obtain the same credit from either lender. A productive advantage in operating collateral expands debt capacity only when the project cannot be transferred without destroying value, so that the specialist recovers more by keeping it than an outside financier recovers by selling it. This makes non-transferability, rather than lender productivity by itself, central to the mechanism.

When such non-transferable value is present, specialized lending changes borrowing and investment even in normal times. Because a specialized lender can recover more from a

[4]That is, the gardener begins drawing down its equity to meet its obligations to creditors.

project in foreclosure, a borrower can credibly promise it a larger repayment; the same project therefore supports more debt, and the borrower needs a smaller equity contribution when new investment opportunities arise. The immediate effect of specialized-lender entry is therefore an expansion in borrowing and investment. The same loans, however, commit more of the project's future payoff to creditors, so the borrower reaches its next investment opportunity with less internal wealth. As highly leveraged borrower vintages accumulate, the initial investment expansion is gradually reversed. In the model's steady state, specialized lending raises debt capacity and leverage per unit of wealth but can reduce borrower land holdings, production, and equilibrium asset prices. The effect of specialized credit is therefore fundamentally dynamic: easier financing today changes borrower wealth accumulation tomorrow.

Consider next an adverse shock to borrower cash flows. If the shock is moderate, specialized lenders absorb the resulting losses by cutting their own payouts while continuing to lend against the going-concern value they preserve, so borrower debt capacity contracts less than in the Kiyotaki–Moore economy. Once intermediary payouts are exhausted, additional losses make internal resources scarce. The intermediary can no longer support the same amount of borrower debt against recovery value that cannot be refinanced externally. Credit capacity then becomes increasingly sensitive to further losses and borrowing conditions tighten. The response to a shock is therefore nonlinear: the marginal effect of an additional unit of shock is small while payouts absorb losses and rises once they are exhausted. This nonlinearity operates through the balance sheet of the specialized lender. It does not by itself make the economy more fragile, since the Kiyotaki–Moore economy also amplifies shocks through collateral values, and it fluctuates more. In our numerical exercises the initial contraction with specialized lenders remains smaller for every shock size we consider, and the economy withstands larger shocks before it collapses. Nonetheless, the trade-off of attenuation is greater persistence of shocks. Borrowers carry more debt out of the downturn, and as the lender's own resources must be rebuilt before it can again support the same borrowing against non-transferable value, investment recovers more slowly. The same lending technology that supports borrowers in normal times generates balance-sheet dependence in stressed states.

We then extend the model to borrowers that differ in pledgeability. Some projects depend heavily on value that only the specialized lender can preserve, as with intangible assets, while

others are backed by collateral that can more readily be transferred to outside financiers, as with tangible assets. In normal times, specialized lending particularly benefits the first group because it allows those firms to borrow against a larger share of their continuation value. The same dependence makes these borrowers the most exposed when the intermediary comes under stress. Loans to low-pledgeability borrowers tie up more of the specialized lender's internal resources, so when those resources become scarce, debt capacity and new originations fall more sharply for these borrowers and lending shifts toward those whose collateral outside financiers can fund. This is the flight to pledgeability. Because both groups are financed from the same balance sheet, the reallocation does not depend on where losses originate: a sufficiently large loss in one part of the lender's portfolio reduces its capacity to finance borrowers elsewhere, including firms whose own fundamentals have not deteriorated. A common intermediary balance sheet creates cross-sector contagion.

Finally, we test the flight-to-pledgeability prediction using business development companies (BDCs), specialized non-bank lenders to middle-market firms, around the COVID-19 shock. BDCs are an empirical counterpart of the specialized lender in the model: they lend against continuation value and carry part of those loans on their own balance sheets. In the model, a shock to the lender's internal resources reduces each borrower's debt capacity in proportion to its pledgeability wedge; a shock to the lender's portfolio should therefore hit less pledgeable borrowers harder. We measure the lender shock by a predetermined exposure to COVID-19, derived from the industry composition of each BDC's portfolio at the end of 2019, and we measure borrower pledgeability by the industry liquidation recovery rates of Kermani and Ma (2023), with higher recovery corresponding to a smaller wedge in the model. Using quarterly PitchBook data on BDC–borrower relationships, we estimate, across BDCs lending to the same borrower, how the cumulative growth of credit from 2019Q4 to a given endpoint varies with lender exposure, and how that relation depends on the recovery rate of the borrower's industry. In the spirit of Khwaja and Mian (2008), the borrower fixed effects absorb the borrower's own demand and performance, so identification comes from differences across lenders within a borrower. Among BDCs lending to the same firm, the more exposed lender cuts credit more, and this gap is largest for borrowers in low-recovery industries: at the end of 2021, a BDC ten percentage points more exposed has a 26-percentage-point more favourable cumulative credit growth for a borrower one standard deviation higher in industry recovery. The interaction exhibits no pre-trend before the shock and fades by 2024.

Literature and context On the theoretical side, our paper is related to the literature on financial amplification through borrower and intermediary balance sheets. In Bernanke et al. (1999) and Kiyotaki and Moore (1997), shocks to borrower net worth or collateral values tighten borrowing constraints and amplify fluctuations. Shleifer and Vishny (1992) emphasize that debt capacity depends on liquidation values, while Rampini and Viswanathan (2010) study the allocation of scarce debt capacity under collateral constraints. A complementary literature places the balance sheet of the intermediary itself at the center of amplification, including Holmstrom and Tirole (1997), He and Krishnamurthy (2013) and Brunnermeier and Sannikov (2014). The closest theoretical antecedent for our model is Rampini and Viswanathan (2019). In both models, intermediaries can support borrowing against value that cannot be fully refinanced with outside capital, so intermediary resources are required for the incremental debt capacity. Our distinction is that this incremental debt capacity arises endogenously as a wedge in recovery values, instead of assuming a fixed difference in collateral enforcement ability. The endogenous recovery value allows the dependence on intermediary capital to vary across assets and states, and we exploit this feature to study heterogeneous pledgeability, endogenous balance-sheet losses, and the resulting reallocation of credit toward more transferable collateral when intermediary resources become scarce.

The predictions of our model can also rationalize several documented facts about private capital markets. **First**, private lenders and private equity sponsors frequently maintain borrower relationships through distress rather than immediately liquidating their claims (Jang, 2024). **Second**, private credit is shown to stabilize the credit cycle (Hinzen et al., 2026) but this may result in slower recoveries due to prolonged distress resolution (Rintamäki and Steffen, 2025). **Third**, direct lenders specialize disproportionately in intangible-intensive firms (Haque et al., 2022), invest heavily in monitoring and industry expertise (Jang et al., 2025; Block et al., 2024). **Fourth**, PE sponsors and direct lenders expertise enables a comparatively higher borrower leverage (Banegas et al., 2026; Chernenko et al., 2022). Our model endogenously generates all these facts consistent with specialized intermediaries creating value through their ability to preserve and finance firms' continuation value rather than simply lending against assets that could be liquidated by any creditor. Our empirical results on flight to pledgeability, further contribute to literature studying BDCs (Davydiuk et al., 2024b,c,a; Chernenko et al., 2025).

Finally, with our focus on the pleagability, our paper is related to the broader corporate

finance literature investigating the distinction between asset-based and cash-flow based debt and its implications to firm financing (Benmelech et al., 2026; Lian and Ma, 2021; Ivashina et al., 2022; Kermani and Ma, 2023). We argue that as firms have experienced a secular shift toward secured debt, as shown by Benmelech et al. (2024), this has created opportunities for private creidt lenders as new types of specialized lenders with expertise in working with firms with little tangeable assets.

Roadmap: The paper proceeds as follows. Section 2 develops the model. Section 3 relates the model to recent evidence on private credit and private equity. Section 4 tests the model's reallocation prediction using BDC lending around COVID-19, and Section 5 concludes.

2 Model

We develop a credit-cycle model in the spirit of Kiyotaki and Moore (1997) to study the role of specialized lenders in the provision of credit. The model introduces a simple form of creditor-specific enforcement. Specialized lenders can preserve part of a borrower's going-concern value following foreclosure, whereas outside lenders can recover only the transferable value of collateral. This recovery advantage expands debt capacity, but requires specialized lenders to commit their own resources. As these resources become scarce, borrowing capacity contracts and shocks can propagate through the lender balance sheet.

2.1 Setting

2.1.1 Environment

Time is discrete, $t = 0, 1, 2, \ldots$. There is a perishable tradable fruit, which is the numeraire, and a fixed aggregate supply K of land. Land trades competitively at price q_t. Land can be bare or cultivated. Cultivated land consists of the underlying land and a stock of trees, which survive from one period to the next at rate $\lambda \in (0, 1)$.

There are three groups of agents. A unit mass of farmers operates cultivated land and borrows against it. A mass $n > 0$ of identical gardeners can hold land and lend to farmers. We interpret gardeners as specialized lenders. A mass $m > 0$ of identical gatherers can hold

land, lend directly to farmers against bare-land collateral, and provide funding to gardeners. Gatherers represent outside financiers who cannot operate the investment.

Let x_t^F, x_t^S, and x_t^G denote consumption by farmers, gardeners, and gatherers. Preferences are linear:

$$U^F = \sum_{t=0}^{\infty} \beta^t x_t^F, \tag{1}$$

$$U^S = \sum_{t=0}^{\infty} (\beta^S)^t x_t^S, \tag{2}$$

$$U^G = \sum_{t=0}^{\infty} (\beta')^t x_t^G, \tag{3}$$

where

$$0 < \beta < \beta^S \leq \beta' < 1. \tag{4}$$

Thus, farmers are more impatient than gardeners, which are weakly more impatient than gatherers.

There is a competitive one-period credit market. Gatherers can lend freely in this market, so their Euler equation pins down the safe gross interest rate:

$$R \equiv \frac{1}{\beta'} \tag{5}$$

Debt is expressed in principal units: principal B_t issued at gross rate R_t at date t requires repayment $R_t B_t$ at date $t+1$.

Within each date, output from assets held at the end of the previous date is realized first and outstanding debt becomes due. Farmers then observe their investment opportunities and choose land holdings, new one-period debt, and consumption. They repay inherited debt using harvest revenue, land sales, and new borrowing. As in Kiyotaki and Moore (1997), farmers supply the labor required for the current harvest before an unanticipated productivity shock. They therefore cannot renegotiate inherited debt by threatening to withdraw that labor. Future labor remains uncommitted, so the human-capital friction continues to limit new and refinanced debt.[5]

[5]See Kiyotaki and Moore (1997), p. 224, footnote 13.

2.1.2 Production and land

Let k_t^F denote cultivated land held by an individual farmer at the end of date t, and let

$$K_t^F \equiv \mathbb{E}[k_t^F] \tag{6}$$

denote aggregate farmer land. Since farmers have unit mass, K_t^F is also average farmer land. Each unit produces $a > 0$ units of tradable fruit and $c > 0$ units of nontradable "bruised fruit." Aggregate tradable output and farmer consumption are therefore

$$Y_{t+1}^F = aK_t^F, \tag{7}$$

$$x_{t+1}^F = cK_t^F. \tag{8}$$

Bruised fruit cannot be traded or pledged.

Let k_t^S and k_t^G denote land held by an individual gardener and gatherer. Their tradable production technologies are $H(k_t^S)$ and $G(k_t^G)$, respectively, with

$$H' > 0, \qquad H'' < 0, \qquad G' > 0, \qquad G'' < 0. \tag{9}$$

Aggregate gardener and gatherer land holdings are nk_t^S and mk_t^G. Land-market clearing requires

$$K_t^F + nk_t^S + mk_t^G = K. \tag{10}$$

Gatherer land demand satisfies

$$q_t - \frac{q_{t+1}}{R} = \frac{1}{R}G'(k_t^G), \tag{11}$$

for an interior landholding. Thus, the standard user cost of land is

$$u_t \equiv q_t - \frac{q_{t+1}}{R} \tag{12}$$

2.1.3 Lumpy investment

As in Section III of Kiyotaki and Moore (1997) - KM97 hereinafter, investment is lumpy. At each date, a farmer receives an investment opportunity with probability $\pi \in (0, 1)$. Only

farmers receiving an opportunity can plant new trees. Planting one unit costs $\phi > 0$ units of tradable fruit.

A farmer without an investment opportunity cannot replace trees that have died. Its noninvestment policy is to retain surviving trees, so it holds

$$k_t^F = \lambda k_{t-1}^F \tag{13}$$

A farmer with an investment opportunity can replace depreciated trees and expand cultivated land. If it ends date t with k_t^F units of cultivated land, investment expenditure is

$$\phi \left(k_t^F - \lambda k_{t-1}^F \right) \tag{14}$$

The stock of trees is specific to the farmer's project. A gardener that takes over the project can continue operating the surviving trees. If cultivated land is instead transferred to a gatherer, the trees are lost and the gatherer receives bare land. Farmers may also sell land to meet inherited obligations, destroying any surviving trees and receiving only its bare-land value, without recovering past planting expenditure. Because investment opportunities are lumpy, these trees cannot necessarily be recreated immediately. This assumption generates non-transferable going-concern value.

Let b_t^F denote the debt position of an individual farmer at the end of date t, with a positive value denoting borrowing and a negative value denoting saving. Aggregate net farmer debt is $B_t^F = \mathbb{E}[b_t^F]$, including borrowing from both lenders and netting out farmer savings.

Each borrowing farmer issues a single one-period claim secured by its land. It borrows directly from a gatherer at R if its entire loan is at most $q_{t+1} k_t^F / R$. Otherwise, a gardener supplies the entire loan and holds its foreclosure rights. Farmers can change lenders freely at renewal and place savings with gatherers.[6]

Let R_t^F be the gross contractual rate on the individual farmer's loan. For $b_t^F > 0$, competitive pricing gives

$$R_t^F = R + \left(\frac{1}{s_t} - R \right) \max \left\{ 1 - \frac{q_{t+1} k_t^F}{R b_t^F}, 0 \right\}, \tag{15}$$

[6]Fully covered loans could instead remain with gardeners, who would then refinance them entirely by borrowing from gatherer. With competitive pricing, free refinancing and full repayment, this leaves aggregate allocations unchanged; see Appendix **??**.

where s_t is the gardener's shadow discount factor defined in (35). For saving and zero positions, we have $R_t^F = R$. Parameters are such that farmers receiving an investment opportunity find it optimal to invest up to the collateral constraint. At the steady state, borrowing up to the collateral constraint pays

$$R^F = \frac{L}{q/R + \beta^S(L-q)}$$

Inherited debt is due at the rate agreed at origination, to the lender that issued the loan, even after an unexpected change in land prices. The farmer budget is therefore:

$$a_t k_{t-1}^F + b_t^F = q_t(k_t^F - k_{t-1}^F) + \phi(k_t^F - \lambda k_{t-1}^F) + R_{t-1}^F b_{t-1}^F \tag{16}$$

Current harvest and new borrowing finance land purchases, planting, and repayment of inherited debt. A farmer without an investment opportunity retains its surviving trees, $k_t^F = \lambda k_{t-1}^F$, and sells the land whose trees have died. Its new debt or saving position is therefore

$$b_t^F = R_{t-1}^F b_{t-1}^F - [a_t + (1-\lambda)q_t]\, k_{t-1}^F. \tag{17}$$

A positive position is financed by gatherers if fully covered by bare land and by gardeners otherwise.

2.2 Specialized lending

The distinction between gardeners and gatherers comes from their recovery values. Let L_{t+1} denote the maximum tradable value that a gardener can recover at date $t+1$ from one unit of cultivated farmer land. Farmer debt is subject to

$$R_t^F b_t^F \leq L_{t+1} k_t^F \tag{18}$$

This constraint binds for farmers receiving an investment opportunity.

Define

$$\chi_t \equiv \frac{L_{t+1}}{R_t^F} \tag{19}$$

as the maximum principal that an investing farmer can borrow per unit of cultivated land

with R_t^F evaluated at the binding collateral constraint. Combining that constraint with (16) gives

$$k_t^F = \frac{(a_t + q_t + \lambda\phi)k_{t-1}^F - R_{t-1}^F b_{t-1}^F}{q_t + \phi - \chi_t} \tag{20}$$

The farmer's required equity contribution per unit of cultivated land is therefore

$$\text{downpayment} \equiv q_t + \phi - \chi_t. \tag{21}$$

Let $B_t^{F,S}$ denote farmer loans held by gardeners and $V_{t+1}^{F,S}$ their transferable value at repayment, anticipated at origination. The difference $B_t^{F,S} - V_{t+1}^{F,S}/R$ is financed with gardener resources. Aggregate farmer repayment, net of returns on farmer savings, is

$$\mathbb{E}[R_t^F b_t^F] = RB_t^F + \left(\frac{1}{s_t} - R\right)\left(B_t^{F,S} - \frac{V_{t+1}^{F,S}}{R}\right). \tag{22}$$

Since investment opportunities are iid, aggregation of the individual investment policy gives

$$\begin{aligned} K_t^F = \; & \underbrace{(1-\pi)\lambda K_{t-1}^F}_{\substack{\text{Land retained by} \\ \text{non-investing farmers}}} \\ & + \underbrace{\frac{\pi}{q_t + \phi - \chi_t}}_{\substack{\text{Investing share} \\ \text{/ down payment}}} \left[\begin{array}{l} \underbrace{(a_t + q_t + \lambda\phi)K_{t-1}^F}_{\text{Assets and harvest}} - \underbrace{RB_{t-1}^F}_{\text{Repayment at } R} \\ - \underbrace{\left(\frac{1}{s_{t-1}} - R\right)\left(B_{t-1}^{F,S} - \frac{V_t^{F,S}}{R}\right)}_{\text{Gardener interest premium}} \end{array} \right] \end{aligned} \tag{23}$$

$$\begin{aligned} B_t^F = \; & \underbrace{RB_{t-1}^F}_{\text{Repayment at } R} + \underbrace{\left(\frac{1}{s_{t-1}} - R\right)\left(B_{t-1}^{F,S} - \frac{V_t^{F,S}}{R}\right)}_{\text{Gardener interest premium}} \\ & + \underbrace{q_t(K_t^F - K_{t-1}^F)}_{\text{Net land purchases}} + \underbrace{\phi(K_t^F - \lambda K_{t-1}^F)}_{\text{Planting expenditure}} - \underbrace{a_t K_{t-1}^F}_{\text{Harvest income}}. \end{aligned} \tag{24}$$

Importantly, the collateral constraint binds for investing farmers, not for the farmer sector

as a whole. As such,

$$\mathbb{E}\left[R_t^F b_t^F\right] \leq L_{t+1} K_t^F \tag{25}$$

with strict inequality in general when $\pi < 1$.

2.2.1 Foreclosure value

A gatherer that forecloses on farmer collateral can recover only the bare-land value q_{t+1}. A gardener can instead operate the surviving trees and liquidate the underlying land as trees die.

A gardener that retains the surviving trees can borrow against their underlying bare land. Retaining λ units therefore supports borrowing of $\lambda q_{t+2}/R$ at date $t+1$, with repayment λq_{t+2} at date $t+2$. Under perfect foresight, the value at date $t+1$ of operating one unit of foreclosed cultivated land is

$$\widehat{L}_{t+1} = (1-\lambda) q_{t+1} + \lambda \left[\frac{q_{t+2}}{R} + s_{t+1} \left(a_{t+2} + L_{t+2} - q_{t+2} \right) \right]. \tag{26}$$

Here $s_{t+1} = \Lambda_{t+2}/\Lambda_{t+1}$ is the gardener's shadow discount factor, defined in (35). Borrowing provides funds at once and is repaid from next period's harvest and continuation value. The value L_{t+2} includes the option to liquidate next period.

The gardener chooses between operating and immediately liquidating the collateral:

$$L_{t+1} = \max \left\{ q_{t+1}, \widehat{L}_{t+1} \right\} \tag{27}$$

Define the pledgeability wedge as

$$W_{t+1} \equiv L_{t+1} - q_{t+1} \geq 0. \tag{28}$$

When $W_{t+1} > 0$, the gardener can preserve value that is lost if collateral passes to a gatherer. We refer to this case as the *operational forceclosure regime*.

In a deterministic steady state with positive gardener consumption, $s = \beta^S$ and

$$L = q + \frac{\lambda \max\{\beta^S a - (1 - 1/R) q, 0\}}{1 - \beta^S \lambda}. \tag{29}$$

Hence, we are in the operational forceclosure regime when

$$\beta^S a > \left(1 - \frac{1}{R}\right) q. \tag{30}$$

The inequality has a simple interpretation. Gardeners preserve the trees when their future harvest is sufficiently valuable relative to bare land.

2.2.2 Gardener balance sheet and loan pricing

Gardeners finance their farmer loan book $B_t^{F,S}$ with their own resources and one-period borrowing from gatherers. Let B_t^S denote aggregate gardener borrowing. Their budget constraint is

$$\begin{aligned} &\underbrace{nq_t(k_t^S - k_{t-1}^S)}_{\text{Net land purchases}} + \underbrace{B_t^{F,S}}_{\text{Farmer lending}} + \underbrace{RB_{t-1}^S}_{\text{Repayment to gatherers}} + \underbrace{x_t^S}_{\text{Consumption}} \\ &= \underbrace{nH(k_{t-1}^S)}_{\text{Own production}} + \underbrace{V_t^{F,S} + \frac{B_{t-1}^{F,S} - V_t^{F,S}/R}{s_{t-1}}}_{\text{Farmer loan repayments}} + \underbrace{B_t^S}_{\text{Borrowing from gatherers}} \end{aligned} \tag{31}$$

Gardeners use resources to purchase land, make farmer loans, repay gatherers, and consume. Their resources come from their own production, repayment of existing farmer loans, and new gatherer borrowing.

If gardeners default, gatherers seize gardener land and the farmer loan book. Gatherers can liquidate gardener land at its market value, but cannot preserve the gardener-specific continuation value embedded in farmer collateral. Their recovery from farmer loans is therefore limited to the bare-land value of the collateral. Summing these collateral values across loans held by gardeners gives $V_{t+1}^{F,S}$. Gardener borrowing is consequently subject to

$$RB_t^S \leq nq_{t+1}k_t^S + V_{t+1}^{F,S}. \tag{32}$$

We focus on the region in which this constraint binds. Appendix A.2 describes the vintage structure used to compute $V_{t+1}^{F,S}$.

Let Λ_t denote the marginal value of one additional unit of tradable resources inside the gardener sector, and let $\xi_t \geq 0$ denote the multiplier on $x_t^S \geq 0$. The gardener consumption

condition is

$$(\beta^S)^t - \Lambda_t + \xi_t = 0, \qquad \xi_t x_t^S = 0, \tag{33}$$

so

$$\Lambda_t = (\beta^S)^t + \xi_t. \tag{34}$$

Define the shadow discount factor

$$s_t \equiv \frac{\Lambda_{t+1}}{\Lambda_t} = \frac{(\beta^S)^{t+1} + \xi_{t+1}}{(\beta^S)^t + \xi_t}. \tag{35}$$

When gardener consumption is positive at both dates, $\xi_t = \xi_{t+1} = 0$ and hence $s_t = \beta^S$.

Let $\mu_t \geq 0$ denote the multiplier on the gardener borrowing constraint (32). The optimality condition for gardener borrowing is

$$\Lambda_t = R\Lambda_{t+1} + R\mu_t. \tag{36}$$

Dividing by Λ_t and using the definition of s_t gives

$$\frac{\mu_t}{\Lambda_t} = \frac{1}{R} - s_t. \tag{37}$$

Gatherers supply fully covered farmer loans at R. Otherwise, the gardener advances b_t^F, borrowing $q_{t+1}k_t^F/R$ against the bare-land claim and providing the remainder with its own resources. It receives $R_t^F b_t^F$ next period and repays $q_{t+1}k_t^F$ to gatherers. Competitive pricing gives

$$b_t^F - \frac{q_{t+1}k_t^F}{R} = s_t \left(R_t^F b_t^F - q_{t+1}k_t^F \right). \tag{38}$$

Solving for the rate gives (15). The contractual rate is a weighted average of R and $1/s_t$, with the weight on R equal to the share of the loan financed against bare land. As this share approaches one, the contractual rate approaches R. Additional borrowing against the same land must be financed entirely with gardener resources, so each extra dollar borrowed adds $1/s_t$ dollars to next period's repayment.

For newly issued farmer loans at the collateral limit, the optimality condition is

$$\Lambda_t = R_t^F \Lambda_{t+1} + \mu_t \frac{q_{t+1}R_t^F}{L_{t+1}}. \tag{39}$$

The last term reflects the additional gatherer funding that one unit of farmer lending can support. A newly originated farmer loan at the collateral limit satisfies $R_t^F b_t^F = L_{t+1} k_t^F$, while gatherers can recover only $q_{t+1} k_t^F$. Hence one additional unit of farmer loan principal increases the pledgeable value of the gardener's loan book by $q_{t+1} R_t^F / L_{t+1}$.

Dividing (39) by $\Lambda_t R_t^F$ and using (37) yields

$$\frac{1}{R_t^F} = s_t + \left(\frac{1}{R} - s_t\right) \frac{q_{t+1}}{L_{t+1}}. \tag{40}$$

Proposition 1 (Gardener debt capacity)**.** *Suppose gardener consumption is positive at both dates, so that $s_t = \beta^S$. The maximum debt capacity of the farmer is*

$$\chi_t = \frac{q_{t+1}}{R} + \beta^S \left(L_{t+1} - q_{t+1}\right) \tag{41}$$

Debt capacity consists of two components. The first, q_{t+1}/R, is borrowing backed by collateral that can be transferred to gatherers. The second, $\beta^S(L_{t+1} - q_{t+1})$, is the additional borrowing capacity generated by the gardener's recovery advantage. Because this recovery advantage is not pledgeable to gatherers, the second component must be financed by gardener resources.

Fully transferable benchmark. If productive assets are fully transferable, a gardener's foreclosure value equals the competitive land price:

$$L_{t+1} = q_{t+1} \tag{42}$$

Equations (40) and (41) then imply

$$R_t^F = R, \qquad \chi_t = \frac{q_{t+1}}{R} \tag{43}$$

The model therefore collapses to the KM97 financing constraint when the gardener has no creditor-specific recovery advantage. Appendix A provides the formal derivation.

Lemma 1 (Neutrality of gardener lending)**.** *When land is fully transferable and no productive capital is lost when ownership changes, gardener lending does not increase farmer debt capacity. Farmers can borrow q_{t+1}/R per unit of land whether they borrow from gardeners or directly from gatherers.*

2.3 Steady state allocation

In an initial steady state without gardeners, farmers borrow from gatherers and can pledge only the transferable value of their land. Their debt capacity is therefore q^0/R per unit of cultivated land. Gardeners can increase farmer debt capacity only if they value seized cultivated land above its liquidation value.

Evaluating equation (29) at the initial land price gives

$$\widehat{L}^0 = \frac{(1-\lambda)q^0 + \beta^S \lambda a}{1-\beta^S \lambda}. \tag{44}$$

Lemma 2 (Gardener entry)**.** *Starting from the steady state without gardener lending, gardeners have a strict recovery advantage if and only if*

$$\beta^S > \frac{q^0}{a+q^0}. \tag{45}$$

The entry condition compares the value of preserving the trees with the value of liquidating the underlying land. When bare land is expensive relative to harvest, gardeners must place a high value on future harvests for continued operation to dominate liquidation. Conversely, when harvest is large relative to the price of bare land, the value preserved by keeping the trees alive is larger.

We first characterize the deterministic steady state in the operational foreclosure regime. When gardener consumption is positive, $s = \beta^S$, the farmer's debt capacity is

$$\chi = \frac{q}{R} + \beta^S (L - q). \tag{46}$$

Lemma 3 (Borrower leverage under gardener finance)**.** *Consider an operational steady state with positive gardener consumption. At a given land price q, an investing farmer financed by gardeners has higher debt relative to output than a farmer financed directly by gatherers:*

$$\frac{\chi}{a} - \frac{q}{aR} = \frac{\beta^S (L-q)}{a} > 0. \tag{47}$$

Gardeners can lend against the continuation value $L - q$ that is lost when collateral passes to a gatherer. They therefore allow the farmers they finance to borrow more per unit

of operating income at the prevailing land price. Stationarity of farmer land requires

$$K^F = (1-\pi)\lambda K^F + \pi \frac{(a+q+\lambda\phi)K^F - R^F B^F}{q+\phi-\chi} \tag{48}$$

The first term is cultivated land carried by farmers without an investment opportunity. The second is the land held after investment by the fraction of farmers able to replant.

The aggregate farmer budget gives

$$(R^F - 1)B^F = [a - (1-\lambda)\phi]\, K^F. \tag{49}$$

Hence,

$$\frac{B^F}{K^F} = \frac{a-(1-\lambda)\phi}{R^F - 1} \tag{50}$$

Since aggregate farmer output is aK^F, equation (50) also gives

$$\frac{B^F}{aK^F} = \frac{a-(1-\lambda)\phi}{a(R^F - 1)}. \tag{51}$$

For given production and planting parameters, aggregate debt relative to output is therefore decreasing in the equilibrium farmer loan rate.

In the operational foreclosure regime, equation (40) implies

$$\frac{1}{R^F} = \beta^S + (\beta' - \beta^S)\frac{q}{L} \leq \beta' = \frac{1}{R}, \tag{52}$$

with strict inequality when $\beta^S < \beta'$. Provided $a > (1-\lambda)\phi$, it follows that

$$\frac{B^F}{aK^F} \leq \frac{a-(1-\lambda)\phi}{a(R-1)}, \tag{53}$$

with equality when $\beta^S = \beta'$ and strict inequality when $\beta^S < \beta'$.

The same operating surplus $a-(1-\lambda)\phi$ services the stationary loan book. A higher loan rate therefore supports less outstanding principal.

Steady-state gardener and gatherer land holdings satisfy

$$q\left(1-\frac{1}{R}\right)=\beta^S H'(k^S), \tag{54}$$

$$q\left(1-\frac{1}{R}\right)=\frac{1}{R}G'(k^G), \tag{55}$$

together with

$$K^F+nk^S+mk^G=K \tag{56}$$

These equations, together with (29), (40), and the gardener balance sheet, characterize the steady-state equilibrium.

Specialized lending has two opposing effects on farmers. On impact, the recovery advantage increases χ and lowers the down payment required to acquire cultivated land. The resulting increase in borrowing, however, raises future debt repayment and lowers the internal resources available when the farmer next receives an investment opportunity. The latter effect can dominate in the stationary distribution.

Proposition 2 (Steady-state effect of gardener lending)**.** *Consider an operational steady state with $L > q$ and positive gardener consumption. Introducing gardener finance into the KM97 benchmark increases farmer debt capacity and leverage for a given land price q but reduces steady-state farmer land K^F, the land price q, farmer consumption x^F, and farmer production Y^F.*

The result reflects the dynamic effect of leverage rather than a contemporaneous contraction in credit. Specialized lending lowers the current down payment and initially expands farmer borrowing, but the associated debt burden reduces future internal resources.

2.4 Credit cycles

We now study the response to an unanticipated one-period decline in farmer productivity. The economy starts from the deterministic steady state in the operational foreclosure regime, so $L^* > q^*$ and $R^{F*} > R$. At date 0,

$$a_0=(1-\Delta)a, \qquad a_t=a \quad \text{for all } t\geq 1, \qquad \Delta\in(0,1). \tag{57}$$

All end-of-date -1 stocks are at their steady-state values.

After date 0, productivity returns immediately to a, but the economy does not return immediately to steady state. The shock changes farmer and gardener balance sheets, and the resulting state variables determine the subsequent path of land prices, borrowing, and investment.

The aggregate farmer equations away from steady state are

$$K_t^F = (1-\pi)\lambda K_{t-1}^F + \pi \frac{(a_t + q_t + \lambda\phi)K_{t-1}^F - R_{t-1}^F B_{t-1}^F}{q_t + \phi - \chi_t}, \tag{58}$$

and

$$B_t^F = R_{t-1}^F B_{t-1}^F + q_t(K_t^F - K_{t-1}^F) + \phi(K_t^F - \lambda K_{t-1}^F) - a_t K_{t-1}^F. \tag{59}$$

Around the steady state, the impact response of farmer land satisfies

$$\left.\frac{dK_0^F}{d\Delta}\right|_{\Delta=0} = \frac{K^{F*}}{q^* + \phi - \chi^*}\left[-(1-\pi)(1-\lambda)\left.\frac{dq_0}{d\Delta}\right|_{\Delta=0} + [1-(1-\pi)\lambda]\left.\frac{d\chi_0}{d\Delta}\right|_{\Delta=0} - \pi a\right]. \tag{60}$$

The shock therefore affects farmer investment directly through current cash flow and indirectly through equilibrium debt capacity.

2.4.1 A small temporary shock

Debt capacity can be written as

$$\chi_t = \frac{q_{t+1}}{R} + s_t(L_{t+1} - q_{t+1}). \tag{61}$$

For comparison, in the KM97 economy,

$$\chi_t^{KM} = \frac{q_{t+1}}{R} \tag{62}$$

The credit-capacity gap is therefore

$$\chi_t - \chi_t^{KM} = s_t(L_{t+1} - q_{t+1}) \geq 0. \tag{63}$$

For sufficiently small shocks, gardener consumption remains positive and $s_t = \beta^S$. Differentiating debt capacity with respect to the shock gives

$$\frac{d\chi_0}{d\Delta} = \frac{1}{R}\frac{dq_1}{d\Delta} + \beta^S\left(\frac{dL_1}{d\Delta} - \frac{dq_1}{d\Delta}\right). \tag{64}$$

In the KM97 benchmark,

$$\frac{d\chi_0^{KM}}{d\Delta} = \frac{1}{R}\frac{dq_1}{d\Delta} \tag{65}$$

The recovery value L_1 depends on the expected path of future land prices and harvests, whereas gatherer recovery depends only on q_1. When

$$\frac{dL_1}{d\Delta} \geq \frac{dq_1}{d\Delta}, \tag{66}$$

the specialized-lender component of debt capacity falls less than the transferable component. Gardener lending then attenuates the standard KM97 collateral channel.

Proposition 3 (Attenuation for small shocks)**.** *Suppose the economy starts in the operational foreclosure regime, gardener consumption remains positive, and condition* (66) *holds. Then farmer debt capacity contracts weakly less after an adverse productivity shock than in the corresponding KM97 economy.*

The attenuation comes from two sources. First, farmers begin with greater debt capacity because $L > q$. Second, the gardener recovery value depends on the continuation value of the project and is therefore less sensitive to a temporary decline in the current liquidation value of land.

Figure 1 illustrates this case. For shocks below the gardener resource threshold, gardener consumption absorbs part of the loss. Farmer debt capacity and land therefore contract less on impact than in the KM97 economy.

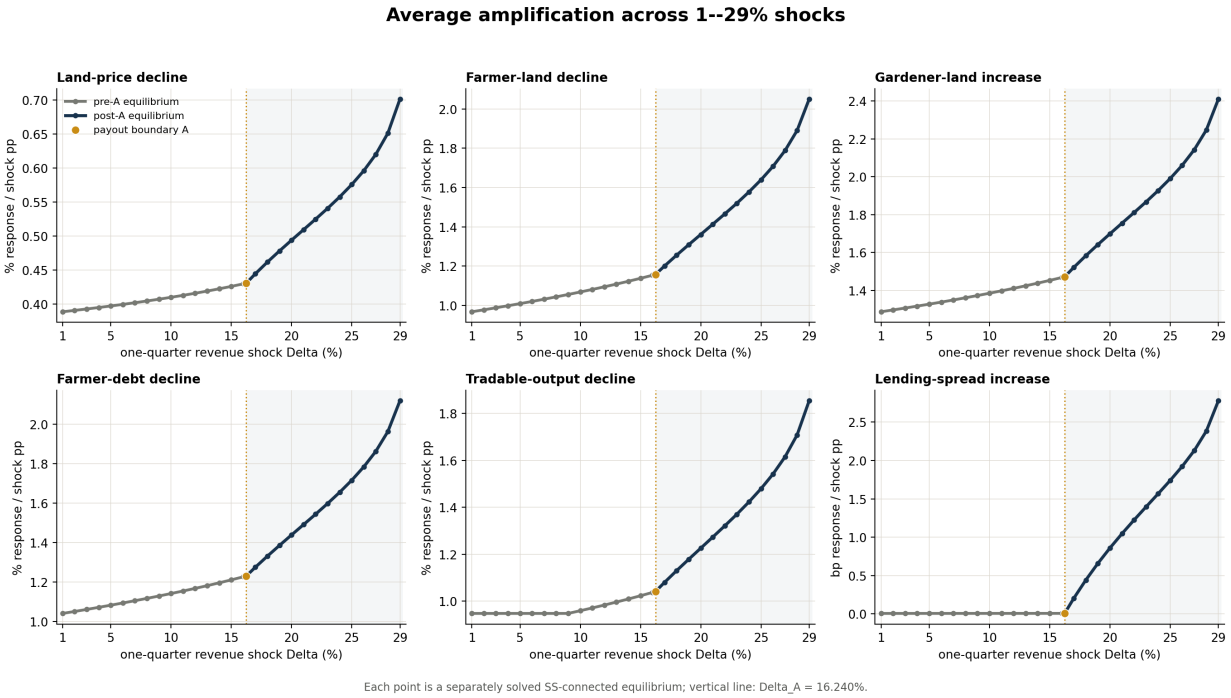


Figure 1: Impact response around gardener consumption buffer exhaustion

Notes: Impact responses as a function of the size of the one-period productivity shock. The vertical line denotes the shock Δ^A at which gardener consumption reaches zero.

2.4.2 Buffer exhaustion

The gardener buffer is finite. Define

$$\Delta^A \equiv \inf\left\{\Delta > 0 : x_0^S(\Delta) = 0\right\} \tag{67}$$

For $\Delta < \Delta^A$, gardeners respond to losses by reducing consumption and $s_0 = \beta^S$. At Δ^A, date-0 gardener consumption reaches zero.

The consumption KKT condition implies

$$s_t = \frac{(\beta^S)^{t+1} + \xi_{t+1}}{(\beta^S)^t + \xi_t} \tag{68}$$

If $x_0^S = 0$ while $x_1^S > 0$, then $\xi_0 > 0$ and $\xi_1 = 0$, so

$$s_0 < \beta^S. \tag{69}$$

Further losses increase the marginal value of current gardener resources and reduce the amount of the non-transferable recovery value that can be supported by the gardener balance sheet.

Allowing s_0 to vary, the response of debt capacity becomes

$$\frac{d\chi_0}{d\Delta} = \frac{1}{R}\frac{dq_1}{d\Delta} + s_0\left(\frac{dL_1}{d\Delta} - \frac{dq_1}{d\Delta}\right) + (L_1 - q_1)\frac{ds_0}{d\Delta} \tag{70}$$

The final term is absent below Δ^A. In the operational foreclosure regime, $L_1 - q_1 > 0$, and after buffer exhaustion $ds_0/d\Delta < 0$. This term therefore creates an additional contraction in debt capacity.

Because

$$R_0^F = \frac{L_1}{\chi_0}, \tag{71}$$

a decline in s_0 also raises the farmer borrowing rate:

$$\frac{\partial R_0^F}{\partial s_0} < 0. \tag{72}$$

Proposition 4 (Buffer exhaustion and nonlinear amplification)**.** *For shocks below Δ^A, gardener consumption adjusts and the shadow discount factor remains at β^S. Once gardener consumption is exhausted, further adverse shocks lower s_0. In the operational foreclosure regime, this decline reduces farmer debt capacity by $(L_1 - q_1)|ds_0|$ and raises the farmer borrowing rate. The response of credit therefore becomes more sensitive to additional losses after Δ^A.*

Figure 1 shows the nonlinear response. In the numerical economy, gardener consumption is exhausted at a shock of approximately 16%. Below this threshold, gardener consumption adjusts to absorb losses. Above it, the marginal value of gardener resources rises and debt capacity becomes more sensitive to the shock as gardener equity is consumed.

Figure 3 shows the same comparison for a 19% shock, above the buffer-exhaustion thresh-

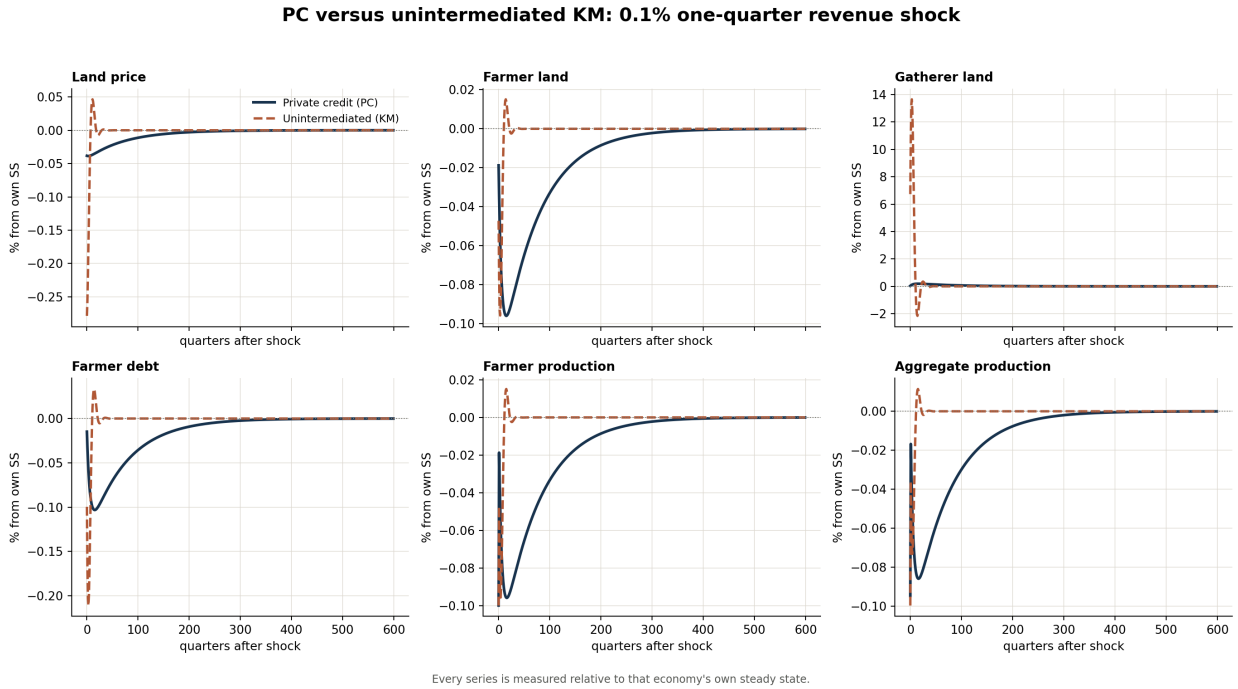


Figure 2: A small temporary shock: gardener lending versus KM

Notes: Responses to a 0.1% one-period decline in farmer tradable harvest, relative to each economy's own steady state. The solid line is the economy with gardener lending; the dashed line is the corresponding KM economy.

old. The gardener economy is now more sensitive to additional losses than it was below Δ^A, but its initial contraction remains smaller than in the KM97 economy. The KM97 economy contracts more sharply and recovers quickly, whereas the gardener economy contracts less on impact and remains below steady state for longer.

2.4.3 Collapse and resilience

Once the consumption buffer is exhausted, the gardener needs to deplete its own resources to fund loans, up to the point the gardening sector cannot afford lending. We can define the collapse threshold as

$$\Delta^C \equiv \inf \{\Delta > 0 : \text{gardeners can no longer meet their inherited obligations}\} \tag{73}$$

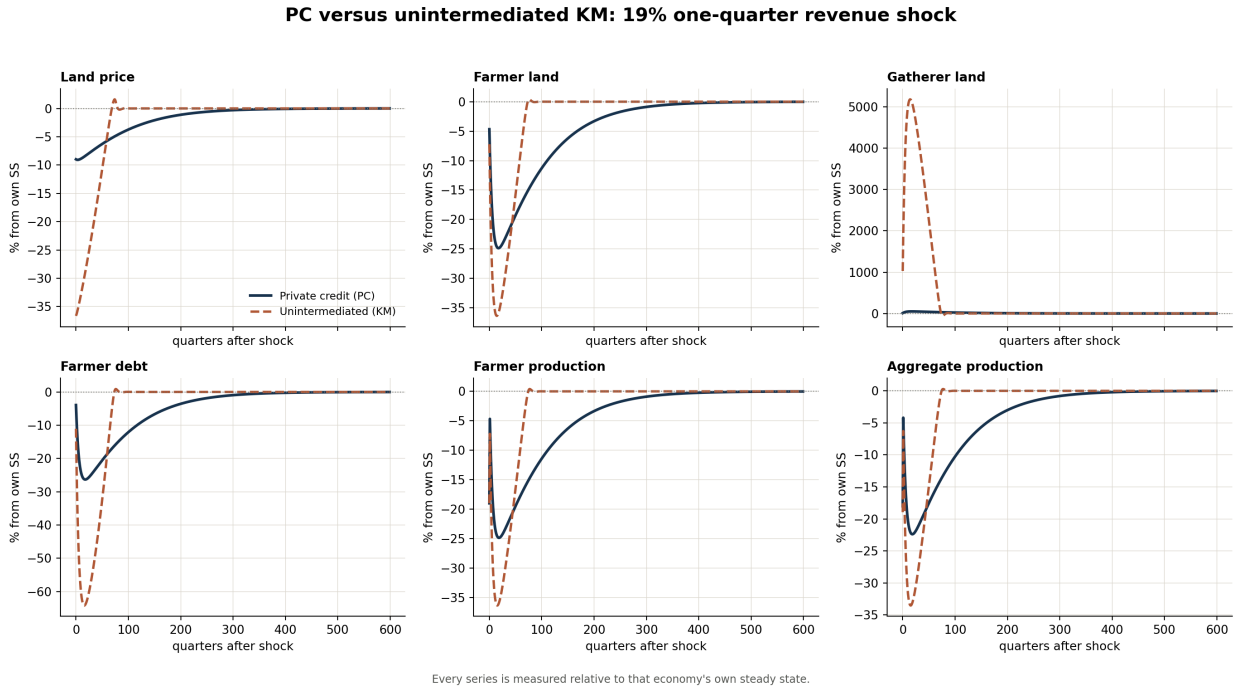


Figure 3: Response after gardener buffer exhaustion

Notes: Perfect-foresight responses to a 19% one-period decline in farmer harvest, which exceeds the gardeners' consumption buffer-exhaustion threshold. All series are measured relative to their own steady-state values.

Define gardener net worth as

$$N_t^S \equiv nq_t k_t^S + B_t^F - B_t^S. \tag{74}$$

When the gardener borrowing constraint binds, combining (32) with (74) gives

$$N_t^S = n\left(q_t - \frac{q_{t+1}}{R}\right) k_t^S + B_t^F - \frac{V_{t+1}^F}{R} \tag{75}$$

The first term is the gardener resources required to support its own land after borrowing against the transferable value of that land. The second is the corresponding own-resource requirement for the farmer loan book. Gardener consumption can therefore reach zero while the sector still has sufficient resources to support a smaller balance sheet.

Lemma 4 (Buffer exhaustion precedes gardener collapse)**.** *For the one-period shock in equa-*

tion (57),

$$\Delta^A \leq \Delta^C. \tag{76}$$

If $x_0^S > 0$ after the shock, gardeners can free additional resources one-for-one by reducing consumption. They therefore cannot fail to meet inherited obligations before this margin is exhausted.

For shocks between Δ^A and Δ^C, gardener consumption is zero and further losses force gardeners to deleverage. Farmer lending falls, gardeners may sell land, and the decline in s_t reduces farmer debt capacity. At Δ^C, deleveraging is no longer sufficient to meet inherited obligations.

In the numerical economy, gardener consumption is exhausted at a shock of about 16%, while gardener collapse occurs at roughly 29%. The corresponding KM97 economy collapses at a shock of about 19%. Thus, stronger amplification after the gardener buffer is exhausted does not imply greater fragility. In this example, gardener lending reduces the initial contraction and allows the economy to withstand a larger shock, at the cost of a slower recovery and lower steady-state production.

2.5 Two farmer sectors: cross-funding and flight to pledgeability

We now allow gardeners to lend to two farmer sectors. The sectors differ in the extent to which the value of cultivated land can be preserved only by gardeners. Because both sectors are financed from the same gardener balance sheet, losses in one sector can affect borrowing in the other.

We call the two farmer types orchardists, indexed by O, and croppers, indexed by C. Orchardists rely more heavily on the continuation value of installed trees, whereas a larger share of cropper collateral value is transferable to gatherers.

2.5.1 A common gardener balance sheet

For each sector $i \in \{O, C\}$, let $(a^i, \lambda^i, \phi^i, \pi^i)$ denote sector-specific production and investment parameters. Each sector has a fixed land supply $\bar{K}^i$ and land price q_t^i. Land-market clearing is

$$K_t^{F,i} + nk_t^{S,i} + mk_t^{G,i} = \bar{K}^i, \qquad i \in \{O, C\} \tag{77}$$

The farmer investment, debt, and recovery equations apply sector by sector. Let L^i_{t+1} denote gardener recovery value from one unit of cultivated type-i farmer land. Debt capacity is

$$\chi^i_t = \frac{q^i_{t+1}}{R} + s_t \left(L^i_{t+1} - q^i_{t+1} \right). \tag{78}$$

The shadow discount factor s_t is common across sectors because both loan books are financed from the same gardener balance sheet.

The gardener resource constraint becomes

$$\begin{aligned} n \sum_{i \in \{O,C\}} q^i_t (k^{S,i}_t - k^{S,i}_{t-1}) + \sum_{i \in \{O,C\}} B^{F,i}_t + R B^S_{t-1} + x^S_t \\ = n \sum_{i \in \{O,C\}} H^i(k^{S,i}_{t-1}) + \sum_{i \in \{O,C\}} R^{F,i}_{t-1} B^{F,i}_{t-1} + B^S_t. \end{aligned} \tag{79}$$

Its borrowing constraint is

$$R B^S_t \leq n \sum_{i \in \{O,C\}} q^i_{t+1} k^{S,i}_t + \sum_{i \in \{O,C\}} V^{F,i}_{t+1} \tag{80}$$

Define the sector-specific pledgeability wedge as

$$W^i_{t+1} \equiv L^i_{t+1} - q^i_{t+1}. \tag{81}$$

We assume

$$W^O_{t+1} > W^C_{t+1} \geq 0. \tag{82}$$

Orchardist credit therefore depends more heavily on gardener resources.

From (78),

$$\frac{\partial \chi^i_t}{\partial s_t} = W^i_{t+1}. \tag{83}$$

Lemma 5 (Differential exposure to gardener resources)**.** *A decline in the shadow discount factor s_t reduces debt capacity in both sectors. The reduction is larger in the sector with the larger pledgeability wedge:*

$$\frac{\partial \chi^O_t}{\partial s_t} > \frac{\partial \chi^C_t}{\partial s_t}. \tag{84}$$

2.5.2 New lending and flight to pledgeability

The same feature that makes gardener lending especially valuable to orchardists in normal times makes them more exposed when gardener resources become scarce. A larger W^i_{t+1} means that a larger part of farmer borrowing is supported by value that cannot be pledged onward to gatherers. When s_t falls, that part of debt capacity contracts.

The effect also carries over to new lending. Let $\mathcal{B}^i_t$ denote the principal issued by the investing farmers in sector i. From the farmer policy rule,

$$\mathcal{B}^i_t = \pi^i \chi^i_t \frac{(a^i_t + q^i_t + \lambda^i \phi^i)\, K^{F,i}_{t-1} - R^{F,i}_{t-1} B^{F,i}_{t-1}}{q^i_t + \phi^i - \chi^i_t}. \tag{85}$$

Holding current farmer resources and land prices fixed, the direct response of new originations to s_t is

$$\left.\frac{\partial \log \mathcal{B}^i_t}{\partial s_t}\right|_{\text{direct}} = \frac{q^i_t + \phi^i}{\chi^i_t\,(q^i_t + \phi^i - \chi^i_t)} W^i_{t+1}. \tag{86}$$

A larger pledgeability wedge reduces new lending through two margins. When gardener resources become scarce, debt capacity falls more, and the resulting increase in the farmer's required equity contribution further reduces the amount of land and debt that can be financed from a given amount of internal wealth.

Under local symmetry across the two sectors except for the pledgeability wedge, the right-hand side of equation (86) is increasing in W^i_{t+1}. A fall in s_t therefore reduces orchardist originations more strongly than cropper originations. The share of new lending going to the more transferable cropper sector rises. We refer to this reallocation as a *flight to pledgeability*.

Proposition 5 (Flight to pledgeability)**.** *Suppose $W^O_{t+1} > W^C_{t+1}$. Once gardener resources become scarce, a decline in s_t reduces orchardist debt capacity more than cropper debt capacity. Conditional on borrower fundamentals, new lending therefore shifts toward the sector with the smaller pledgeability wedge.*

2.5.3 Cross-sector transmission

Consider an unanticipated one-period productivity shock to croppers only:

$$a^C_0 = (1-\Delta)a^C, \qquad a^C_t = a^C \quad \text{for } t \geq 1, \qquad a^O_t = a^O \quad \text{for all } t. \tag{87}$$

Orchardist productivity is therefore unchanged.

For shocks small enough that gardener consumption remains positive, the common shadow discount factor remains at β^S. The cropper shock affects orchardists only through equilibrium prices and the common gardener balance sheet. Once the shock exhausts gardener consumption, however,

$$\frac{ds_0}{d\Delta} < 0. \tag{88}$$

The effect of this common balance-sheet contraction on sector-i debt capacity is

$$W_1^i \frac{ds_0}{d\Delta} \tag{89}$$

The differential effect across sectors is therefore

$$\left(W_1^O - W_1^C\right) \frac{ds_0}{d\Delta} < 0. \tag{90}$$

Proposition 6 (Cross-sector transmission)**.** *Suppose $W_1^O > W_1^C$ and a shock to cropper productivity is sufficiently large to exhaust gardener consumption. The resulting decline in s_0 reduces orchardist debt capacity even though orchardist productivity is unchanged. Through this common balance-sheet channel, the high-wedge sector can contract more than the sector directly hit by the productivity shock.*

Figure 4 illustrates this transmission mechanism. Although only cropper productivity falls, orchardist land prices fall more, and the contractions in orchardist farmer land and debt are substantially larger. Land released by orchardist farmers is absorbed by gardeners and gatherers, whose orchardist land holdings rise by more than their cropper holdings.

The orchardist contraction does not reflect a deterioration in orchardist productivity. Rather, orchardist borrowing relies more heavily on the gardener-specific component of collateral value. The cropper shock exhausts gardener consumption and lowers s_0 for both sectors. Because $W^O > W^C$, the resulting contraction in debt capacity is larger for orchardists.

Thus, the reallocation of credit need not follow the location of the underlying productivity shock. In Figure 4, croppers are the only farmers whose productivity falls, yet scarce gardener resources shift financing away from orchardists because orchardist loans require more specialized-lender balance-sheet support.

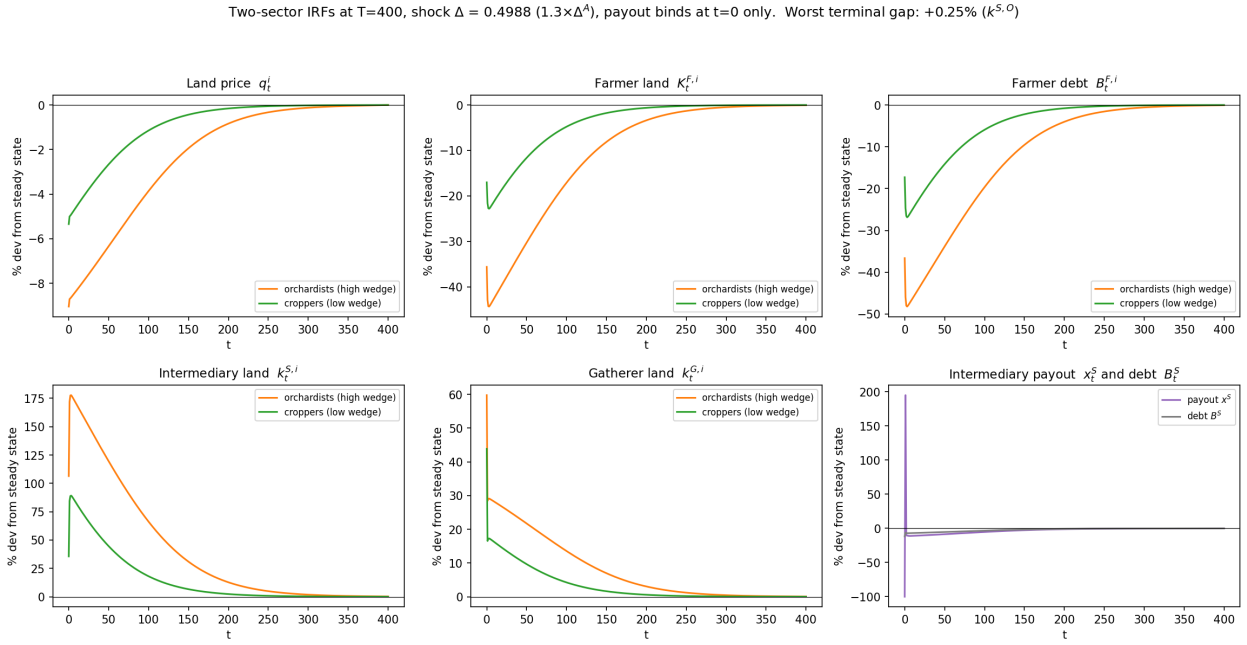


Figure 4: Cross-sector transmission of a cropper shock

Notes: Perfect-foresight responses to a one-quarter decline in cropper harvest. Orchardist productivity is unchanged. Orchardists are the high-wedge sector and croppers the low-wedge sector. The shock is $\Delta = 0.4988$, approximately 1.3 times the buffer-exhaustion threshold in this two-sector economy, and gardener consumption is constrained at date 0 only. All series are measured relative to their own steady-state values.

The same mechanism applies to a shock that directly reduces gardener resources. More generally, any disturbance that lowers s_t reduces sector-i debt capacity in proportion to its pledgeability wedge W^i_{t+1}. This mechanism provides a balance-sheet channel through which losses in one part of a specialized lender's portfolio can affect credit supply to otherwise unrelated borrowers.

3 Facts about private capital intermediaries

We apply our model to understand the persistent growth of the private credit industry. Translating our agrarian agents into modern market participants, the farmers are the borrowing firms. The gardeners are the private credit lenders and the private equity sponsors behind them, who have the expertise to operate portfolio companies that unsophisticated

lenders and investors lack. Finally, the gatherers are the rest of the savers in the economy, who do not have the expertise to run the portfolio firms independently. In this section, we highlight four recently established facts about private credit that our model can explain. These facts correspond, in order, to four propositions of the model.

Fact 1: Direct lenders specialize in firms with intangible assets and invest heavily in monitoring and industry expertise. Dell'Ariccia et al. (2021) show that the substantial growth in intangible capital since the mid-1980s explains a large fraction of the secular decline in the share of commercial lending in banks' loan portfolios. Private credit, on the other hand, is disproportionately important for private equity-backed firms in intangible-intensive industries. Jang et al. (2025) find that the expansion of direct lending has been concentrated in enterprise-value-based lending to PE-backed firms operating in intangible-capital-intensive industries. Direct lenders write collateral claims that place greater weight on the continuation value of the firm following default, particularly in intangible industries, and are substantially more specialized by industry than banks.

Direct lenders also invest heavily in monitoring. Block et al. (2024); Jang (2024) document extensive use of financial and incurrence covenants by private debt funds. These findings suggest that the distinction between private credit and more arm's-length lending is more than the form of collateral securing the loan. Direct lenders combine concentrated ownership, information acquisition, covenants, monitoring, and industry-specific expertise in financing borrowers whose value may be difficult to realize through a simple asset sale.

As shown in Proposition 1, gardners have superior debt capacity than gatherers due to their recovery advantage. Such advantage is greatest for firms whose continuation value is difficult to transfer to outside creditors, hence rationalizing private credit's specialization.

Fact 2: Private credit borrowers tend to be more leveraged than borrowers in the syndicated loan market. Chernenko et al. (2022) show that direct lenders lend to less profitable and more leveraged firms than the traditional broadly syndicated loans. Similarly, Banegas et al. (2026) document: 5x Debt/EBITDA-ratio for private credit borrowers versus 3.2x for broadly syndicated loan borrowers, and lower interest coverage ratio for private credit borrowers.

As shown in lemma 3 and Proposition 2, the model directly generates higher borrower leverage because specialized lenders can lend against both transferable collateral and additional lender-specific continuation value, allowing borrowers to support more debt against

the same underlying project.

Fact 3: Private credit financing is less cyclical than traditional leveraged lending. Private credit, or "direct lending", becomes relatively more important during adverse credit-market conditions. Hinzen et al. (2026) document direct lending market is less cyclical than leveraged loan market. As a result, the borrowers switch to direct lending from syndicated lending when credit conditions tighten or when they experience distress.

As shown by Proposition 3, the model generates this fact because specialized lenders can lend against going-concern value when transferable collateral values fall, cushioning the decline in borrowers' debt capacity following moderate adverse shocks.

Fact 4: Private lenders and sponsors maintain borrower relationships through distress. Direct loans are typically privately negotiated and held by the originating lender rather than distributed through an active secondary market. Because many private credit instruments lack a liquid secondary market, lenders commonly hold loans until maturity or refinancing. This ownership structure and lack of observable market prices can facilitate bilateral renegotiation when borrowers encounter financial difficulties. Recent literature has shown that this flexibility may help but sometimes make recognizing distress more gradual (Jang, 2024; Rintamäki and Steffen, 2025)[7]

The evidence on private equity-backed borrowers shows a similar preference for maintaining the firm as a going concern. Hotchkiss et al. (2021, 2023); Haque et al. (2022) find that distressed PE-backed firms restructure more quickly, are more likely to avoid bankruptcy court, and are less likely to liquidate than otherwise comparable highly leveraged firms. Sponsors are also more likely to retain control following restructuring and frequently inject additional capital as firms approach distress.

Proposition 4 shows that, for sufficiently small shocks, gardener consumption adjusts to sustain lending to farmers during distress. The model thus rationalizes the empirical fact because specialized lenders preserve borrowers' going-concern value that would otherwise be lost in liquidation.

[7]A related development in private equity is the growing use of continuation vehicles, which allow sponsors to extend their ownership of portfolio companies beyond the typical life of the original fund (Ivashina et al., 2026; Abuzov et al., 2025) Thus, longer holding periods might allow investors to "ride out" the bad times, but this can make the recovery times also more prolonged.

4 Empirical evidence: COVID-19 flight to pledgeability

The two-sector model makes a sharp theoretical prediction: once gardener resources become scarce, credit contracts most in the sector with the larger pledgeability wedge (Lemma 5 and Proposition 5). In that sector, a larger share of borrowing depends on recovery value that the gardener cannot pledge to gatherers. We test this prediction using business development companies (BDCs) portfolios around COVID-19.

4.1 Data and measurement

BDCs are specialized non-bank lenders to middle-market firms. Their quarterly filings disclose their portfolios borrower by borrower in Schedules of Investments, which PitchBook compiles into security-level holdings. Our sample runs from 2018Q1 through 2024Q1. We retain debt positions with positive reported principal, exclude equity positions, and aggregate securities to the BDC–borrower–quarter level. Before the endpoint-specific restrictions described below, the resulting network contains 72 BDCs, 2,947 borrowers financed by at least two BDCs at some point during the sample, and 9,947 realized BDC–borrower links. The credit measure $B_{\ell bt}$ is principal outstanding, in millions of dollars. Fair value is used only to construct the lender exposure measure below.

We measure lender exposure to the COVID shock using the industry composition of each BDC's portfolio in 2019Q4. For each PitchBook IndustryCode k, let

$$C_k = -100 \frac{GO_{k,2020} - GO_{k,2019}}{GO_{k,2019}}, \tag{91}$$

denote the contraction in annual BEA gross output from 2019 to 2020, after mapping the PitchBook industry classification to BEA industries. A larger C_k denotes a more adverse industry shock. BDC ℓ's exposure is

$$E_\ell = \frac{\sum_{j \in \mathcal{P}_{\ell,-1}} FV_{\ell j,-1} C_{k(j)}}{\sum_{j \in \mathcal{P}_{\ell,-1}} FV_{\ell j,-1}}, \tag{92}$$

where $\mathcal{P}_{\ell,-1}$ is the BDC's debt portfolio in 2019Q4 and $FV_{\ell j,-1}$ is the fair value of position j. Thus, E_ℓ is a 2019Q4 portfolio-weighted exposure to industries that contracted during COVID-19, measured in percentage points of output contraction. The portfolio weights are

fixed before COVID-19, so subsequent lending cannot alter the exposure measure. Positions without an industry classification remain in the denominator and contribute zero to the numerator. E_ℓ measures exposure to COVID-hit industries.

The exposure sample contains 72 BDCs with positive debt portfolios in all four quarters of 2019, of which 69 enter at least one endpoint regression. Across the 72 BDCs, E_ℓ has a mean of -0.25 percentage points, a standard deviation of 3.90, and a 10th-to-90th percentile range of -4.47 to 3.24. A ten-percentage-point exposure difference, the scale used below, is therefore equal to 2.57 standard deviations.

We measure borrower pledgeability using the midpoint property, plant, and equipment recovery rate in Chapter 11 from Kermani and Ma (2023). The source measure is reported by two-digit SIC industry. We map each PitchBook borrower to its SIC industries, average the recovery rates when a borrower has more than one matched SIC industry, and then take the unweighted mean across matched borrowers within each PitchBook IndustryGroup. The borrower–SIC crosswalk contains 12,478 borrowers, of which 3,681 have at least one matched SIC industry and 731 have more than one. The resulting IndustryGroup measure covers 35 groups and is standardized with equal weight across those groups. Its unstandardized mean is 34.36 percent of book value and its standard deviation is 7.17 percentage points. We denote the standardized measure by $\widetilde{R}_b$. A higher $\widetilde{R}_b$ means that more collateral value is transferable in liquidation and hence that the pledgeability wedge is smaller. Table 1 reports the main summary statistics.

Table 1: Summary statistics

Variable	Observations	Mean	S.D.	10th percentile	Median	90th percentile
BDC COVID exposure (percentage points)	72	−0.25	3.90	−4.47	−0.43	3.24
IndustryGroup liquidation recovery (percent)	35	34.36	7.17	26.31	34.51	44.11
2019Q4 relationship principal ($ millions)	4,266	12.68	25.67	0.00	4.18	30.67
2021Q4 relationship principal ($ millions)	4,266	15.64	32.81	0.00	4.71	39.00
Cumulative symmetric principal growth (percentage points)	4,266	1.74	166.72	−200.00	−0.92	200.00

Notes: BDC exposure is summarized across lenders with positive debt portfolios in all four quarters of 2019. Liquidation recovery is summarized across the 35 PitchBook IndustryGroups with a mapped Kermani–Ma recovery rate. The last three rows use the fitted 2021Q4 endpoint sample after the borrower fixed-effect restriction. A zero principal at one endpoint denotes relationship entry or exit, which leads to +200 or -200 growth, respectively.

4.2 Cumulative credit-growth design

We index quarters relative to 2020Q1: $h = -1$ is the 2019Q4 baseline, $h = 0$ is 2020Q1, $h = 3$ is 2020Q4, $h = 7$ is 2021Q4, and $h = 16$ is 2024Q1. For each endpoint h, a BDC–borrower pair enters the cross-section if the BDC's full debt portfolio is observed with positive total principal in both 2019Q4 and the endpoint quarter and if $B_{\ell b,h} + B_{\ell b,-1} > 0$. This retains relationships that enter or exit between the two dates and excludes pairs with zero principal at both dates. Borrower fixed effects further require at least two usable BDC observations for a borrower in a given endpoint cross-section.

The dependent variable is the Davis–Haltiwanger–Schuh symmetric cumulative growth rate,

$$g_{\ell bh} = 100 \frac{2\,(B_{\ell b,h} - B_{\ell b,-1})}{B_{\ell b,h} + B_{\ell b,-1}}. \tag{93}$$

It equals +200 percentage points for an entering relationship and −200 percentage points for an exiting relationship. Because $B_{\ell bt}$ is principal outstanding, the outcome combines new lending and add-ons with amortization, repayment, maturity, and relationship entry or exit. It therefore measures the net adjustment in outstanding credit rather than new originations directly.

For each endpoint h, we estimate

$$g_{\ell bh} = \gamma_h \left(\frac{E_\ell}{10}\right) + \beta_h \left(\frac{E_\ell}{10}\right) \widetilde{R}_b + \mu_{bh} + \varepsilon_{\ell bh}, \tag{94}$$

where μ_{bh} is a borrower fixed effect in endpoint h. The fixed effect structure restricts the comparison to BDCs lending to the same borrower at the same endpoint and absorbs borrower-level changes common across those lenders. The coefficient γ_h is the relative credit response to a ten-percentage-point difference in lender exposure for a borrower with mean recovery. The coefficient β_h measures how that response changes when borrower-industry recovery is one standard deviation higher.

We estimate equation (94) by weighted least squares. The weight is the relationship's baseline-endpoint midpoint share of the BDC's debt portfolio,

$$w_{\ell bh} = \frac{B_{\ell b,h} + B_{\ell b,-1}}{A_{\ell h} + A_{\ell,-1}}, \qquad A_{\ell h} = \sum_{b'} B_{\ell b',h}, \tag{95}$$

where $A_{\ell h}$ is BDC ℓ's full observed debt-portfolio principal at endpoint h. Standard errors are two-way clustered by BDC and borrower. Exposure varies only across BDCs, of which between 55 and 65 enter a given endpoint regression, and its distribution has a long right tail, so a handful of lenders carry much of the identifying variation. Inference at the BDC level therefore uses restricted, null-imposed, two-sided wild-cluster bootstrap tests with BDC bootstrap weights and the same BDC–borrower two-way studentization (Cameron et al., 2008; MacKinnon and Webb, 2017), and we report these bootstrap p-values throughout.

4.3 Results

Table 2 reports estimates at three endpoints. At the end of 2021, the lower-order exposure coefficient is $\widehat{\gamma}_7 = -40.59$ percentage points ($p = 0.005$): a more exposed BDC reduced credit relative to a less exposed BDC lending to the same borrower with mean industry recovery. The interaction coefficient is $\widehat{\beta}_7 = 25.86$ percentage points ($p = 0.007$). Thus, for two BDCs whose COVID exposures differ by ten percentage points, a one-standard-deviation increase in borrower-industry recovery makes the relative cumulative credit response 25.86 percentage points more favorable. This is a relative reallocation result. At one standard deviation above mean recovery, the implied exposure effect remains negative at -14.73 percentage points, so the estimate means that exposed BDCs cut credit less to higher-recovery borrowers, in line with our numerical simulations and flight-to pledgeability.

As the model predicts, the interaction is insignificant at the end of 2020, peaks over 2021 and 2022, and fades by 2024 as lender balance sheets recover. Figure 5 shows both coefficients at every endpoint. There is no pre-trend: the seven pre-COVID coefficients are individually small and jointly insignificant ($p = 0.723$).

4.4 Robustness

The main identification concern is that lender exposure may partly reflect the BDC's holdings in the focal borrower's own industry. To remove this mechanical overlap, we construct a leave-own-industry-out exposure for each BDC–borrower observation. It excludes every 2019Q4 position in the borrower's IndustryGroup from both the numerator and denominator of equation (92). The 2021Q4 interaction is then 26.09 percentage points ($p = 0.017$), compared with 25.86 in the baseline specification. The estimate is also similar when principal rather

Table 2: Within-borrower cumulative credit growth from 2019Q4

	2020Q4 $h = 3$	2021Q4 $h = 7$	2024Q1 $h = 16$
Exposure × recovery, β_h	7.05	25.86***	10.42
	(8.35)	(9.33)	(11.58)
BDC-WCB p-value	[0.418]	[0.007]	[0.361]
Exposure, γ_h	−30.06***	−40.59***	−10.43
	(11.64)	(13.01)	(23.21)
BDC-WCB p-value	[0.002]	[0.005]	[0.802]
Observations	4,022	4,266	4,556
BDCs	65	58	55
Borrowers	1,387	1,501	1,566
IndustryGroups	33	32	32
Borrower fixed effects	Yes	Yes	Yes
Midpoint portfolio-share weights	Yes	Yes	Yes

Notes: Each column estimates equation (94) in a separate endpoint cross-section. The dependent variable is cumulative symmetric growth in BDC–borrower principal outstanding from 2019Q4 to the stated endpoint, in percentage points. The variable h denotes quarters relative to 2020Q1, so $h = -1$ is the 2019Q4 baseline. Exposure is divided by ten and recovery is standardized across IndustryGroups. Parentheses report standard errors two-way clustered by BDC and borrower. Brackets report restricted, null-imposed, two-sided wild-cluster-bootstrap p-values using 9,999 Rademacher replications, BDC bootstrap weights, and BDC–borrower two-way studentization. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

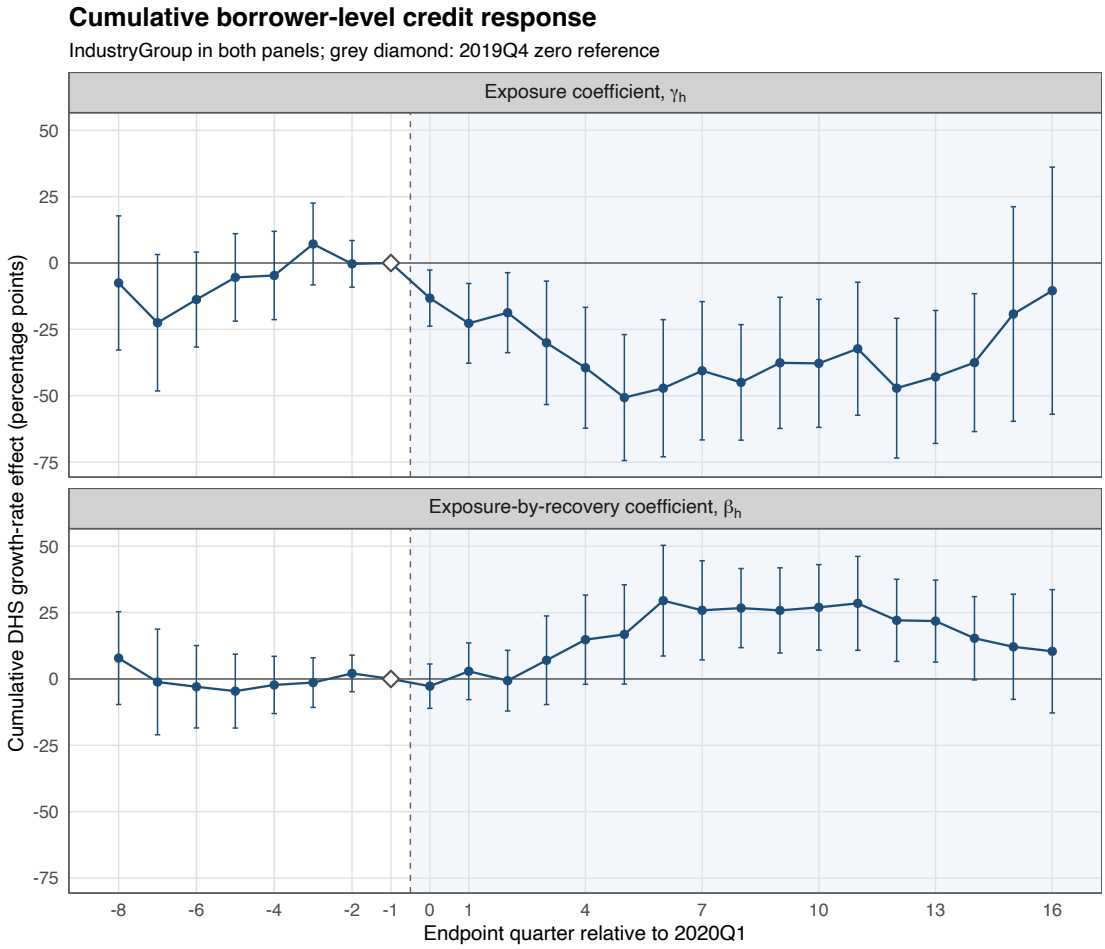


Figure 5: Cumulative within-borrower coefficients using IndustryGroup recovery. The top panel plots the exposure coefficient $\widehat{\gamma}_h$ and the bottom panel plots the exposure-by-recovery coefficient $\widehat{\beta}_h$ from equation (94). The dependent variable is symmetric growth in principal outstanding between 2019Q4 and endpoint h, where h denotes quarters relative to 2020Q1. The diamond at $h = -1$ is the mechanically zero 2019Q4 reference. Both panels show one specification, use midpoint portfolio-share weights, and report 95 percent confidence intervals based on standard errors two-way clustered by BDC and borrower.

than fair value weights the 2019Q4 portfolio exposure. Appendix Figure 8 shows why the correction is small: the industries that generate exposure are high-recovery industries, so own-industry deleveraging would push the interaction toward zero.

Appendix B reports the complete robustness results and alternative recovery measures. A partial adjustment for principal reaching contractual maturity preserves a positive interaction of 20.37 percentage points, but with weaker precision ($p = 0.074$). The estimate remains positive when recovery is aggregated at the finer PitchBook IndustryCode level and when it is assigned directly from matched SIC industries. It is smaller and imprecise under equal regression weights and among relationships with positive principal at both endpoints. The reallocation therefore operates largely through relationship entry and exit rather than through the size of continuing loans, as shown in appendix B.

5 Conclusion

This paper develops a credit-cycle model in which specialized lenders can preserve borrower going-concern value that arm's-length financiers cannot fully realize. This recovery advantage expands debt capacity and helps stabilize credit after moderate shocks, but it also makes lending increasingly dependent on intermediary balance-sheet capacity. When specialized-lender resources become scarce, credit contracts more sharply and shifts toward borrowers with more transferable collateral. The BDC evidence around COVID is broadly consistent with this flight-to-pledgeability mechanism. Overall, the paper shows how specialized lending can both support credit provision and create a distinct channel of financial amplification.

Our results help to understand the transmission of shocks in the economy where specialized intermediaries play a role. The implications of our results extend to the recent financing trends in private credit. One is that private credit lenders are increasingly engaging in asset based lending, such as investing in data centers that are collateralized with tangible and hard collateral. Our flight to pledgability mechansim illustrates that if these tangible assets investments trigger a large enough shock that influences lender's balance sheet, this initial shock could trigger an even bigger response in the—originally unaffected—less pledgeable sector.

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A Additional Model Derivations

This appendix provides additional derivations for the model in Section 2. We first derive the fully transferable benchmark and the pledgeable value of the gardener's farmer loan book. We then study the introduction of gardener lending into an economy initially at the steady state without gardener lending.

A.1 Fully transferable productive assets

Consider first an economy in which land is the only productive asset and is fully transferable. Because land is homogeneous and can be transferred without destroying any productive capital, the same unit of land can move freely between farmers, gardeners, and gatherers. Differences in productivity across the three groups therefore affect the competitive land price q_t rather than creating distinct physical collateral values.

Consider one unit of farmer land seized by a gardener at date $t+1$. The gardener has two options. It can sell the land immediately for q_{t+1}, or it can retain the land and operate it.

Let $\mu_t \geq 0$ denote the multiplier on the gardener funding constraint (32), normalized by the marginal value of gardener resources. Dividing the gardener borrowing condition by the marginal value of resources gives $\mu_t/\Lambda_t = 1/R - s_t$. For an interior gardener landholding, the first-order condition for land is

$$q_t = s_t \left[H'(k_t^S) + q_{t+1}\right] + (1/R - s_t)q_{t+1} \tag{96}$$

The left-hand side is the cost of one additional unit of land at date t. The first term on the right-hand side is its value to gardeners through production and resale at date $t+1$. The

second term reflects the additional gatherer borrowing that the land can support.

Now consider the value, at date $t+1$, of retaining one unit of land obtained through foreclosure rather than selling it. Denote this value by $\widetilde{L}_{t+1}$. From the same gardener problem,

$$\widetilde{L}_{t+1} = s_{t+1}\left[H'(k^S_{t+1}) + q_{t+2}\right] + (1/R - s_{t+1})q_{t+2} \tag{97}$$

Keep in mind that when gardener consumption is positive at both t and $t+1$, its resources aren't scarce and $s_t = \beta^S$. Such is the case in steady state.

Equation (96), evaluated one period ahead, implies

$$\widetilde{L}_{t+1} = q_{t+1} \tag{98}$$

whenever gardeners hold land in equilibrium. The recovery value of farmer collateral to a gardener is therefore

$$L_{t+1} = \max\left\{q_{t+1}, \widetilde{L}_{t+1}\right\} = q_{t+1}. \tag{99}$$

This equality is the central result of the benchmark. A gardener may be a more productive user of land than a gatherer, but that does not create a creditor-specific recovery value. Because land is fully transferable, its productive value is already reflected in its competitive market price. A gatherer that receives the land can sell it back into that market for the same q_{t+1}.

The farmer collateral constraint consequently becomes

$$R^F_t B^F_t \leq q_{t+1} K^F_t. \tag{100}$$

Gatherers can lend directly against exactly this collateral at the safe rate R. Competition

therefore implies

$$R_t^F = R. \tag{101}$$

Substituting equations (99) and (101) into the definition of debt capacity gives

$$\chi_t = \frac{q_{t+1}}{R}. \tag{102}$$

Thus, despite the presence of gardeners, the benchmark steady state coincides with the KM97 steady state in farmer production and the equilibrium land price. Because land is fully transferable, foreclosure does not create any creditor-specific recovery value: both gardeners and gatherers can ultimately realize the same market value q_t from the collateral.

A.2 Pledgeable value of the farmer loan book

Let $K_t^{F,j}$ and $B_t^{F,j}$ denote the aggregate land and net debt of farmers whose last investment occurred j periods ago. All farmers investing at date t borrow χ_t per unit of land. Between investment opportunities, harvest, land sales, and repayments are proportional to their land holdings, so farmers within each vintage continue to have the same debt-to-land ratio. Aggregate farmer land and net debt are

$$K_t^F = \sum_{j \geq 0} K_t^{F,j}, \qquad B_t^F = \sum_{j \geq 0} B_t^{F,j}. \tag{103}$$

Gardeners finance vintages whose borrowing exceeds bare-land debt capacity. Their loan book and the collateral value they can pledge to gatherers are therefore

$$B_t^{F,S} = \sum_{\substack{j\geq 0: \\ B_t^{F,j} > q_{t+1}K_t^{F,j}/R}} B_t^{F,j}, \tag{104}$$

$$V_{t+1}^{F,S} = q_{t+1} \sum_{\substack{j\geq 0: \\ B_t^{F,j} > q_{t+1}K_t^{F,j}/R}} K_t^{F,j}. \tag{105}$$

Farmers in the remaining vintages borrow from or save with gatherers. The part of the gardener loan book that cannot be financed against bare land requires gardener resources:

$$B_t^{F,S} - \frac{V_{t+1}^{F,S}}{R} = \sum_{j\geq 0} \max\left\{ B_t^{F,j} - \frac{q_{t+1}K_t^{F,j}}{R}, 0 \right\}. \tag{106}$$

Summing repayments on gardener loans and direct gatherer loans, net of returns on farmer savings, gives

$$\begin{aligned} \mathbb{E}[R_t^F b_t^F] &= V_{t+1}^{F,S} + \frac{B_t^{F,S} - V_{t+1}^{F,S}/R}{s_t} + R(B_t^F - B_t^{F,S}) \\ &= RB_t^F + \left(\frac{1}{s_t} - R\right)\left(B_t^{F,S} - \frac{V_{t+1}^{F,S}}{R}\right). \end{aligned} \tag{107}$$

The first two terms on the first line are repayments to gardeners; the last is repayment to gatherers net of returns on farmer savings. The interest premium applies only to the principal financed with gardener resources.

Since investment opportunities are iid, the newly investing vintage holds

$$K_t^{F,0} = \pi \frac{(a_t + q_t + \lambda\phi)K_{t-1}^F - RB_{t-1}^F - (1/s_{t-1} - R)\left(B_{t-1}^{F,S} - V_t^{F,S}/R\right)}{q_t + \phi - \chi_t}, \tag{108}$$

$$B_t^{F,0} = \chi_t K_t^{F,0}. \tag{109}$$

A share $1-\pi$ of each previous vintage receives no new investment opportunity. These farmers retain the fraction λ of land with surviving trees and use harvest and land-sale proceeds to repay debt. Along an anticipated price path, their land and net debt satisfy, for $j \geq 1$,

$$K_t^{F,j} = (1-\pi)\lambda K_{t-1}^{F,j-1}, \tag{110}$$

$$\begin{aligned} B_t^{F,j} = {} & (1-\pi)\left[RB_{t-1}^{F,j-1} - \{a_t + (1-\lambda)q_t\}K_{t-1}^{F,j-1}\right] \\ & + (1-\pi)\left(\frac{1}{s_{t-1}} - R\right)\max\left\{B_{t-1}^{F,j-1} - \frac{q_t K_{t-1}^{F,j-1}}{R}, 0\right\}. \end{aligned} \tag{111}$$

At the unexpected date-0 shock, inherited repayments were agreed using the steady-state price q and discount factor β^S. The debt equation is therefore

$$\begin{aligned} B_0^{F,j} = {} & (1-\pi)\left[RB_{-1}^{F,j-1} - \{a_0 + (1-\lambda)q_0\}K_{-1}^{F,j-1}\right] \\ & + (1-\pi)\left(\frac{1}{\beta^S} - R\right)\max\left\{B_{-1}^{F,j-1} - \frac{qK_{-1}^{F,j-1}}{R}, 0\right\}. \end{aligned}$$

The land equation is unchanged. Current harvest and land sales reflect the shock; inherited repayments remain fixed.

For the newly investing vintage,

$$R_t^F B_t^{F,0} = L_{t+1}K_t^{F,0}. \tag{112}$$

When $L_{t+1} > q_{t+1}$, financing this vintage requires $s_t(L_{t+1} - q_{t+1})K_t^{F,0}$ of gardener resources.

Gardener net worth is

$$N_t^S \equiv nq_t k_t^S + B_t^{F,S} - B_t^S. \tag{113}$$

Gatherers lend to gardeners and fully covered farmers, and accept farmer savings. Their budget constraint is

$$\begin{aligned} mq_t(k_t^G - k_{t-1}^G) + B_t^S + B_t^F - B_t^{F,S} + x_t^G \\ = mG(k_{t-1}^G) + R\left(B_{t-1}^S + B_{t-1}^F - B_{t-1}^{F,S}\right). \end{aligned} \tag{114}$$

In the steady state considered in the main text, gardener consumption is positive, so

$$s = \beta^S. \tag{115}$$

Proof of Lemma ??. In the operational foreclosure regime, $W = L - q > 0$. With positive gardener consumption, $s = \beta^S$. As long as vintage j borrows from gardeners, its budget gives

$$\frac{B^{F,j+1}}{K^{F,j+1}} = q - \frac{a}{\lambda} + \frac{B^{F,j}/K^{F,j} - q/R}{\lambda\beta^S}. \tag{116}$$

Using the recovery value in (29) and the initial debt-to-land ratio $B^{F,0}/K^{F,0} = q/R + \beta^S W$ gives

$$\frac{B^{F,j}}{K^{F,j}} = \frac{q}{R} + \beta^S\left[W + a - a(\lambda\beta^S)^{-j}\right]. \tag{117}$$

Debt per unit of land falls with vintage age. Let J be the first vintage whose loan is fully covered by bare land. The formula above remains valid at J, so the cutoff satisfies

$$(\lambda\beta^S)^J \leq \frac{a}{a+W} < (\lambda\beta^S)^{J-1}. \tag{118}$$

Vintage J borrows or saves at the safe rate, with repayment per unit of land

$$\frac{RB^{F,J}}{K^{F,J}} = q - R\beta^S \left[a(\lambda\beta^S)^{-J} - a - W\right] \leq q \tag{119}$$

All subsequent vintages also borrow or save at R. For $j \geq J$,

$$\frac{RB^{F,j}}{K^{F,j}} = \frac{R[a+(1-\lambda)q]}{R-\lambda} + \left(\frac{R}{\lambda}\right)^{j-J} \left[\frac{RB^{F,J}}{K^{F,J}} - \frac{R[a+(1-\lambda)q]}{R-\lambda}\right]. \tag{120}$$

The operational condition implies $q < R[a+(1-\lambda)q]/(R-\lambda)$. Debt per unit of land therefore continues to fall. Farmers borrow from gatherers while debt remains positive and save with gatherers thereafter, earning R.

These policies allow farmers to repay inherited debt at each date:

$$a + (1-\lambda)q + \lambda\chi - L = a(1-\lambda\beta^S) > 0. \tag{121}$$

The first three terms are harvest revenue, sales of land whose trees have died, and borrowing capacity against surviving cultivated land. Together they exceed L, the largest inherited repayment per unit of land.

Aggregate lending and saving. In steady state, a vintage loses the share π of farmers who invest again and the share $1-\lambda$ of land whose trees die. Its share of farmer land is

therefore

$$\frac{K^{F,j}}{K^F} = \left[1-(1-\pi)\lambda\right]\left[(1-\pi)\lambda\right]^j. \tag{122}$$

Only vintages $0,\ldots,J-1$ borrow from gardeners. Summing their collateral, loan principal, and repayments gives

$$V^{F,S} = qK^F\left\{1-\left[(1-\pi)\lambda\right]^J\right\}, \tag{123}$$

$$B^{F,S} = \sum_{j=0}^{J-1} K^{F,j}\left\{\frac{q}{R}+\beta^S\left[W+a-a(\lambda\beta^S)^{-j}\right]\right\}, \tag{124}$$

$$RB^{F,S}+\left(\frac{1}{\beta^S}-R\right)\left(B^{F,S}-\frac{V^{F,S}}{R}\right) = \sum_{j=0}^{J-1} K^{F,j}\left[L+a-a(\lambda\beta^S)^{-j}\right]. \tag{125}$$

Gardener resources committed to lending, per unit of farmer land, are therefore

$$\frac{B^{F,S}-V^{F,S}/R}{K^F} = \beta^S\left[1-(1-\pi)\lambda\right]\sum_{j=0}^{J-1}\left[(1-\pi)\lambda\right]^j\left[W+a-a(\lambda\beta^S)^{-j}\right]. \tag{126}$$

For a fixed cutoff J, this expression is affine in W and hence in the land price. Appendix A.3 uses this property to show that the stationary farmer equations have at most one land price.

The remaining vintages borrow from or save with gatherers. Their aggregate net repayment is finite when $(1-\pi)R<1$: the share of farmers who have not reinvested must decline faster than their savings grow. Under this condition,

$$\begin{aligned}\frac{R(B^F-B^{F,S})}{K^F} = {}& \left[1-(1-\pi)\lambda\right]\left[(1-\pi)\lambda\right]^J \\ & \times\left[\frac{R[a+(1-\lambda)q]}{(R-\lambda)[1-(1-\pi)\lambda]}+\frac{RB^{F,J}/K^{F,J}-R[a+(1-\lambda)q]/(R-\lambda)}{1-(1-\pi)R}\right].\end{aligned} \tag{127}$$

Dividing by R gives their net principal per unit of farmer land. Aggregate direct gatherer lending and farmer savings are, respectively,

$$\sum_{\substack{j \geq J: \\ B^{F,j}>0}} B^{F,j}, \qquad \sum_{\substack{j \geq J: \\ B^{F,j}>0}} B^{F,j} - \left(B^F - B^{F,S}\right). \tag{128}$$

The first sum is finite because sufficiently old vintages become savers. Both totals follow from (120) and the land shares in (122)

Note that in the benchmark with fully transferable land, $L = q$ and farmers borrow from gatherers from the outset, so $B^{F,S} = V^{F,S} = 0$. Investing farmers borrow q/R per unit of land. Their subsequent debt follows (120), setting $J = 0$ and $B^{F,0}/K^{F,0} = q/R$.

A.3 Gardener entry

To better understand how the introduction of a gardening sector affects the equilibrium, we now study the introduction of gardener lending into an economy initially at the steady state without gardener lending. Let q^0 denote the land price in this initial steady state. We first compare stationary economies in which gardeners finance different shares of new farmer investment. We then study how the additional leverage created by gardener finance affects farmer wealth after origination.

A.3.1 Entry at the initial steady state

At the initial steady state, farmers borrow from gatherers and can pledge only the transferable value of their land. Their debt capacity is therefore q^0/R per unit of cultivated land.

Gardeners can increase farmer debt capacity only if they value seized cultivated land above its liquidation value.

Evaluating equation (29) at the initial land price gives

$$\widehat{L}^0 = \frac{(1-\lambda)q^0 + \beta^S \lambda a}{1 - \beta^S \lambda}. \tag{129}$$

Continued operation is strictly more valuable than liquidation when $\widehat{L}^0 > q^0$. Rearranging gives

$$\beta^S > \frac{q^0}{a + q^0}. \tag{130}$$

Proposition 7 (Gardener entry)**.** *Starting from the steady state without gardener lending, gardeners have a strict recovery advantage if and only if*

$$\beta^S > \frac{q^0}{a + q^0}. \tag{131}$$

The entry condition compares the value of preserving the trees with the value of liquidating the underlying land. When bare land is expensive relative to harvest, gardeners must place a high value on future harvests for continued operation to dominate liquidation. Conversely, when harvest is large relative to the price of bare land, the value preserved by keeping the trees alive is larger.

When condition (130) holds, let

$$L^0 \equiv \widehat{L}^0 > q^0 \tag{132}$$

denote gardener recovery value at the initial steady state.

For the remainder of this subsection, we impose

$$\beta^S = \beta' = \frac{1}{R}. \tag{133}$$

Gardeners and gatherers then value future tradable fruit in the same way. Any difference between gardener and gatherer finance therefore comes from the additional recovery value $L^0 - q^0$, rather than from differences in discounting.

To describe the resources of an investing farmer, define

$$e_t^F \equiv (a + q_t + \lambda\phi)\, k_{t-1}^F - R_{t-1}^F b_{t-1}^F. \tag{134}$$

This is the numerator of the farmer policy in equation (20): current harvest, the value of inherited land, and the replacement cost saved on surviving trees, net of inherited debt repayment. Consider a farmer receiving an investment opportunity at the initial steady state with internal resources $e_t^F = e > 0$.

Under gatherer finance, debt capacity is q^0/R. The farmer's required equity contribution per unit of cultivated land, as defined in equation (21), is therefore

$$q^0 + \phi - \frac{q^0}{R}. \tag{135}$$

Under gardener finance, equation (41) and the equal-discount restriction (133) imply

$$\chi = \frac{L^0}{R}. \tag{136}$$

The required farmer equity contribution therefore falls to

$$q^0 + \phi - \frac{L^0}{R}. \tag{137}$$

We assume this quantity is positive. Since $L^0 > q^0$,

$$q^0 + \phi - \frac{L^0}{R} < q^0 + \phi - \frac{q^0}{R}. \tag{138}$$

The farmer consequently acquires

$$\frac{e}{q^0 + \phi - \frac{L^0}{R}} > \frac{e}{q^0 + \phi - \frac{q^0}{R}} \tag{139}$$

units of cultivated land under gardener finance rather than gatherer finance. More precisely, the increase is

$$\frac{e}{q^0 + \phi - \frac{L^0}{R}} - \frac{e}{q^0 + \phi - \frac{q^0}{R}} = \frac{\frac{e}{R}\left(L^0 - q^0\right)}{\left(q^0 + \phi - \frac{q^0}{R}\right)\left(q^0 + \phi - \frac{L^0}{R}\right)} > 0. \tag{140}$$

Gardener finance therefore increases land purchases by a farmer receiving an investment opportunity at the prevailing price. The farmer begins with the same internal resources, but can finance a larger fraction of the acquisition and replanting cost with debt.

A.3.2 Stationary share of gardener lending

We next compare stationary economies in which gardener lending accounts for different shares of new farmer investment. Let $\theta \in [0, 1]$ denote the fraction of farmers receiving an investment opportunity whose new borrowing is financed by gardeners in every period. The

remaining fraction $1-\theta$ borrows from gatherers. Assignment of the two sources of finance is independent of farmers' current internal resources. Thus θ indexes a sequence of stationary economies; it is not a transition path.

Under the equal-discount restriction (133), both gardener-financed and gatherer-financed farmer loans carry the gross rate R. Aggregating farmer budgets across both sources of finance therefore gives the same steady-state debt-service identity as equation (49). Per unit of farmer land, outstanding principal is

$$\frac{a-(1-\lambda)\phi}{R-1}. \tag{141}$$

At a stationary land price q, farmers financed by gatherers contribute

$$q+\phi-\frac{q}{R} \tag{142}$$

of their own resources per unit of new cultivated land. Farmers financed by gardeners contribute

$$q+\phi-\frac{L(q)}{R}, \tag{143}$$

where, in the operational foreclosure regime,

$$L(q)=\frac{(1-\lambda)q+\frac{\lambda}{R}a}{1-\frac{\lambda}{R}}. \tag{144}$$

Stationarity of farmer land requires

$$1-(1-\pi)\lambda=\pi\left[a+q+\lambda\phi-R\frac{a-(1-\lambda)\phi}{R-1}\right]\times\left[\frac{1-\theta}{q+\phi-\frac{q}{R}}+\frac{\theta}{q+\phi-\frac{L(q)}{R}}\right]. \tag{145}$$

The left-hand side is the fraction of farmer cultivated land that must be replenished in each period. The first bracket on the right-hand side gives the internal resources of farmers receiving an investment opportunity per unit of inherited farmer land. The second converts these resources into newly cultivated land, taking into account that gardener-financed farmers require less equity per unit of land.

When $\theta=0$, equation (145) reduces to the stationary condition without gardener lending and gives $q=q^0$.

Suppose the initial steady state satisfies

$$\pi>\left(1-\frac{1}{R}\right)\left[1-(1-\pi)\lambda\right]. \tag{146}$$

This is the local regularity condition for the stationary equilibrium branch through $(q^0,0)$. Differentiating equation (145) with respect to θ at $\theta=0$ gives

$$\left.\frac{dq}{d\theta}\right|_{\theta=0}=-\frac{\left[1-(1-\pi)\lambda\right]\left(q^0+\phi-\frac{q^0}{R}\right)}{\pi-\left(1-\frac{1}{R}\right)\left[1-(1-\pi)\lambda\right]}\times\left[\frac{q^0+\phi-\frac{q^0}{R}}{q^0+\phi-\frac{L^0}{R}}-1\right]. \tag{147}$$

At $\theta=0$, changes in $L(q)$ induced by changes in q do not enter this derivative: they affect only the gardener-financed term in equation (145), whose weight is zero at the point

of differentiation.

Since $L^0 > q^0$,

$$\frac{q^0 + \phi - \frac{q^0}{R}}{q^0 + \phi - \frac{L^0}{R}} > 1. \tag{148}$$

Condition (146) therefore implies

$$\left.\frac{dq}{d\theta}\right|_{\theta=0} < 0. \tag{149}$$

Proposition 8 (Stationary share of gardener lending)**.** *Suppose gardeners have a strict recovery advantage at the initial steady state, farmer equity contributions remain positive, and condition* (146) *holds. Along the stationary equilibrium branch through the initial steady state,*

$$\left.\frac{dq}{d\theta}\right|_{\theta=0} < 0. \tag{150}$$

A marginal increase in the stationary share of new farmer investments financed by gardeners therefore lowers the stationary land price.

This result contrasts with the individual response in equation (139). At the initial price, access to gardener finance allows an investing farmer to purchase more land. Yet a stationary economy in which gardener finance is used more frequently has a lower land price. The reason is that the additional borrowing changes the farmer's wealth after the initial investment.

A.3.3 Leverage displacement

Under the equal-discount restriction (133), an investing farmer financed directly by gatherers borrows q/R per unit of cultivated land. A farmer financed by gardeners borrows $L(q)/R$.

Their debt-to-output ratios at origination are therefore

$$\frac{q}{aR} \qquad \text{and} \qquad \frac{L(q)}{aR}. \tag{151}$$

Since $L(q) > q$ in the operational foreclosure regime,

$$\frac{q/(aR)}{L(q)/(aR)} = \frac{q}{L(q)} < 1. \tag{152}$$

Using equation (144),

$$\frac{d}{dq}\left(\frac{q}{L(q)}\right) = \frac{\lambda a}{R\left(1-\frac{\lambda}{R}\right)L(q)^2} > 0. \tag{153}$$

It follows from Proposition 8 that

$$\frac{d}{d\theta}\left(\frac{q}{L(q)}\right)\bigg|_{\theta=0} = \frac{\lambda a}{R\left(1-\frac{\lambda}{R}\right)(L^0)^2}\,\frac{dq}{d\theta}\bigg|_{\theta=0} < 0. \tag{154}$$

A marginal increase in the share of gardener finance therefore lowers direct origination leverage relative to gardener-financed origination leverage.

Both financing-specific leverage ratios decline when the land price falls, but direct leverage is more sensitive to q. Indeed,

$$0 < \frac{d}{dq}\left(\frac{L(q)}{aR}\right) = \frac{1-\lambda}{aR\left(1-\frac{\lambda}{R}\right)} < \frac{1}{aR} = \frac{d}{dq}\left(\frac{q}{aR}\right). \tag{155}$$

Consequently,

$$\frac{d}{d\theta}\left(\frac{q}{aR}\right)\bigg|_{\theta=0} < \frac{d}{d\theta}\left(\frac{L(q)}{aR}\right)\bigg|_{\theta=0} < 0. \tag{156}$$

The decline in the stationary land price reduces the borrowing capacity of both groups, but it reduces the borrowing capacity of directly financed farmers by more.

The same displacement appears in new borrowing. Per unit of farmer internal resources, new principal equals

$$\frac{q/R}{q+\phi-q/R} \qquad \text{under direct gatherer finance,} \tag{157}$$

and

$$\frac{L(q)/R}{q+\phi-L(q)/R} \qquad \text{under gardener finance.} \tag{158}$$

The ratio of the first quantity to the second is

$$\frac{q}{L(q)}\frac{q+\phi-L(q)/R}{q+\phi-q/R}<1. \tag{159}$$

The first factor is increasing in q by equation (153). The second factor also satisfies

$$\frac{d}{dq}\left[\frac{q+\phi-L(q)/R}{q+\phi-q/R}\right]=\frac{\frac{\lambda}{R}\left(1-\frac{1}{R}\right)\left(\phi+\frac{a}{R}\right)}{\left(1-\frac{\lambda}{R}\right)\left(q+\phi-\frac{q}{R}\right)^2}>0. \tag{160}$$

Hence,

$$\frac{d}{d\theta}\left[\frac{q}{L(q)}\frac{q+\phi-L(q)/R}{q+\phi-q/R}\right]\Bigg|_{\theta=0}<0. \tag{161}$$

Proposition 9 (Leverage displacement)**.** *Suppose the conditions of Proposition 8 hold. A marginal increase in the stationary share of gardener finance lowers debt relative to output for both gatherer-financed and gardener-financed investing farmers, but lowers it by more under gatherer finance. New principal per unit of farmer internal resources also falls under gatherer finance relative to gardener finance.*

Gardener finance therefore reallocates leverage toward the farmers it finances. As its stationary share rises, the land-price decline compresses borrowing against transferable collateral more strongly than borrowing supported by gardener recovery value.

Under the equal-discount restriction, this displacement is compositional rather than aggregate. Equation (141) implies that total outstanding farmer principal relative to output is

$$\frac{a-(1-\lambda)\phi}{a(R-1)}, \tag{162}$$

independently of θ. When $\beta^S < \beta'$, gardener loans instead carry a higher rate and equation (53) shows that aggregate debt relative to output is lower.

A.3.4 Leverage and subsequent farmer wealth

Consider again a farmer with internal resources e that receives an investment opportunity at the initial steady state. Hold the land price at q^0 and compare the farmer's resources at its next investment opportunity under the two financing arrangements.

Under gatherer finance, the farmer purchases

$$\frac{e}{q^0+\phi-\frac{q^0}{R}} \tag{163}$$

units of cultivated land. Its debt repayment is q^0 per unit. Its internal resources at the next investment opportunity are therefore

$$e_1^G = (a+\lambda\phi)\,\frac{e}{q^0+\phi-\frac{q^0}{R}}. \tag{164}$$

Under gardener finance, the farmer purchases

$$\frac{e}{q^0 + \phi - \frac{L^0}{R}} \tag{165}$$

units of cultivated land and repays L^0 per unit. Its internal resources at the next investment opportunity are therefore

$$e_1^S = \left[a + \lambda\phi - \left(L^0 - q^0\right)\right] \frac{e}{q^0 + \phi - \frac{L^0}{R}}. \tag{166}$$

Subtracting equations (164) and (166) gives

$$e_1^S - e_1^G = \frac{e\left(L^0 - q^0\right)}{\left(q^0 + \phi - \frac{q^0}{R}\right)\left(q^0 + \phi - \frac{L^0}{R}\right)} \left[\frac{a + \lambda\phi}{R} - \left(q^0 + \phi - \frac{q^0}{R}\right)\right]. \tag{167}$$

The initial steady-state condition implies

$$\begin{aligned} q^0 + \phi - \frac{q^0}{R} - \frac{a + \lambda\phi}{R} \\ = \frac{\frac{1}{R}\left(1 - \frac{1}{R}\right)\left[1 - (1 - \pi)\lambda\right](a + \lambda\phi)}{\pi - \left(1 - \frac{1}{R}\right)\left[1 - (1 - \pi)\lambda\right]}. \end{aligned} \tag{168}$$

Under condition (146), the right-hand side is positive. Hence

$$e_1^S < e_1^G. \tag{169}$$

Proposition 10 (Gardener finance and farmer wealth)**.** *Starting from the initial steady state, a farmer financed by gardeners acquires more land at its current investment opportunity but reaches its next investment opportunity with less internal wealth than an otherwise identical*

farmer financed by gatherers.

Gardener finance has opposite effects at the current and subsequent investment opportunities. The additional recovery value lowers the farmer's required equity contribution today and allows it to purchase more land. But this expansion is financed by a larger claim on the future payoff from that land. The farmer consequently carries less internal wealth into its next investment opportunity.

Increasing θ creates newly gardener-financed farmers that purchase more land at origination, but it also increases the mass of farmers whose past gardener borrowing has left them with less internal wealth. A stationary equilibrium reflects both effects.

A.3.5 A one-time cohort experiment

The propagation of this wealth effect can be isolated by a one-time cohort experiment. Start from the initial steady state and hold the land price fixed at q^0. At date 0, give a marginal cohort of investing farmers access to gardener finance, while otherwise identical farmers receive gatherer finance. From date 1 onward, both groups again use gatherer finance whenever they receive an investment opportunity. The experiment therefore changes financing only once and deliberately holds prices fixed; it is not the equilibrium transition following gardener entry.

Let ΔK_t^F denote the difference in farmer land attributable to this one-time change in financing. On impact,

$$\Delta K_0^F = \frac{\frac{e}{R}\left(L^0 - q^0\right)}{\left(q^0 + \phi - \frac{q^0}{R}\right)\left(q^0 + \phi - \frac{L^0}{R}\right)} > 0. \tag{170}$$

The cohort therefore initially holds more land. At subsequent investment opportunities,

however, equation (169) implies that the same farmers have fewer internal resources with which to acquire and replant land.

Solving the fixed-price vintage recursion gives

$$\Delta K_t^F = \Delta K_\infty^F + \left(\Delta K_0^F - \Delta K_\infty^F\right)\left[R(1-\pi)\lambda\right]^t, \tag{171}$$

provided

$$R(1-\pi)\lambda < 1. \tag{172}$$

The long-run cohort effect satisfies

$$\Delta K_\infty^F < 0. \tag{173}$$

Thus the farmers receiving gardener finance at date 0 initially hold more land than their gatherer-financed counterparts, but their land advantage declines as the cohort ages and eventually reverses. The extra debt capacity expands their balance sheets at origination; the associated loss of retained wealth subsequently limits how much cultivated land they can finance when new investment opportunities arrive.

Newly gardener-financed farmers contribute positively to farmer land demand, while older gardener-financed vintages contribute negatively through their lower accumulated wealth. Proposition 8 shows that, locally around the initial steady state, the latter force is strong enough for a greater stationary share of gardener lending to lower the equilibrium land price.

A.3.6 A numerical entry path

The preceding cohort experiment holds prices fixed to isolate the effect of gardener finance on farmer wealth. Figure 6 shows the corresponding equilibrium transition when gardeners enter with a small initial equity position and gradually expand their lending.

Starting from a small balance sheet, gardeners accumulate equity and finance a progressively larger share of new farmer investment. Gardener-financed farmer land expands as the stock of farmers financed by gatherers is gradually replaced. The land price, however, begins to fall well before aggregate farmer land adjusts substantially. This is the forward-looking price effect described above: the price reflects the lower future internal wealth of the increasingly leveraged farmer population before that loss of wealth has fully worked its way through the distribution of farmer vintages. As gardener-financed vintages become dominant, aggregate farmer land falls toward its lower steady-state level and the land released by farmers is absorbed by gardeners and gatherers. Oscillations in land stock are caused by the different farmer vintages' wealth. Gardener equity and the stock of gardener-financed farmers temporarily overshoot their long-run levels during this transition before converging.

A.4 Additional derivations for the response to shocks

For completeness, this subsection provides additional expressions underlying the credit-cycle results in Section 2.4.

For the one-period shock considered in the main text, farmer debt capacity is

$$\chi_t = \frac{q_{t+1}}{R} + s_t \left(L_{t+1} - q_{t+1} \right). \tag{174}$$

Differentiating with respect to the size of the shock gives

$$\frac{d\chi_0}{d\Delta} = \left(\frac{1}{R} - s_0\right)\frac{dq_1}{d\Delta} + s_0\frac{dL_1}{d\Delta} + (L_1 - q_1)\frac{ds_0}{d\Delta}. \tag{175}$$

In the corresponding KM economy, farmer debt capacity is simply $\chi_0^{KM} = q_1/R$. Hence

$$\frac{d\chi_0}{d\Delta} - \frac{d\chi_0^{KM}}{d\Delta} = s_0\left(\frac{dL_1}{d\Delta} - \frac{dq_1}{d\Delta}\right) + (L_1 - q_1)\frac{ds_0}{d\Delta}. \tag{176}$$

For shocks that leave gardener consumption positive at dates 0 and 1, $ds_0/d\Delta = 0$ and $s_0 = \beta^S$, so

$$\frac{d\chi_0}{d\Delta} - \frac{d\chi_0^{KM}}{d\Delta} = \beta^S\left(\frac{dL_1}{d\Delta} - \frac{dq_1}{d\Delta}\right). \tag{177}$$

Since the primitive productivity shock has disappeared at date 1,

$$L_1 = (1-\lambda)q_1 + \beta^S\lambda\,(a + L_2) \tag{178}$$

in the operational foreclosure regime.

Once gardener consumption is exhausted and s_0 falls with the size of the shock, equation (176) contains the additional negative term

$$(L_1 - q_1)\frac{ds_0}{d\Delta} < 0. \tag{179}$$

Since $R_0^F = L_1/\chi_0$, holding L_1 and q_1 fixed,

$$\frac{\partial R_0^F}{\partial s_0} = -\frac{L_1(L_1 - q_1)}{\chi_0^2} < 0. \tag{180}$$

These expressions make explicit the two components of the comparison with the KM economy. Before buffer exhaustion, the response differs because gardener recovery value can be less sensitive to the shock than bare-land value. After buffer exhaustion, the decline in the shadow value of future gardener resources introduces the additional balance-sheet term $(L_1 - q_1)ds_0/d\Delta$.

A.5 Additional derivation for new lending across sectors

For an investing farmer in sector i, new debt principal is $\chi_t^i k_t^{F,i}$. Aggregating across the fraction π^i of farmers receiving an investment opportunity gives

$$\mathcal{B}_t^i = \pi^i \chi_t^i \frac{\left(a_t^i + q_t^i + \lambda^i \phi^i\right) K_{t-1}^{F,i} - R_{t-1}^{F,i} B_{t-1}^{F,i}}{q_t^i + \phi^i - \chi_t^i}. \tag{181}$$

Holding current farmer resources and land prices fixed, the direct response of new originations to s_t is

$$\left.\frac{\partial \log \mathcal{B}_t^i}{\partial s_t}\right|_{\text{direct}} = \frac{q_t^i + \phi^i}{\chi_t^i \left(q_t^i + \phi^i - \chi_t^i\right)} W_{t+1}^i. \tag{182}$$

Under local symmetry across the two sectors except for the pledgeability wedge, the right-hand side of equation (182) is increasing in W_{t+1}^i. A fall in s_t therefore reduces orchardist originations more strongly than cropper originations. The share of new lending going to the more transferable cropper sector rises. This is the flight to pledgeability described in Section 2.5. "‘

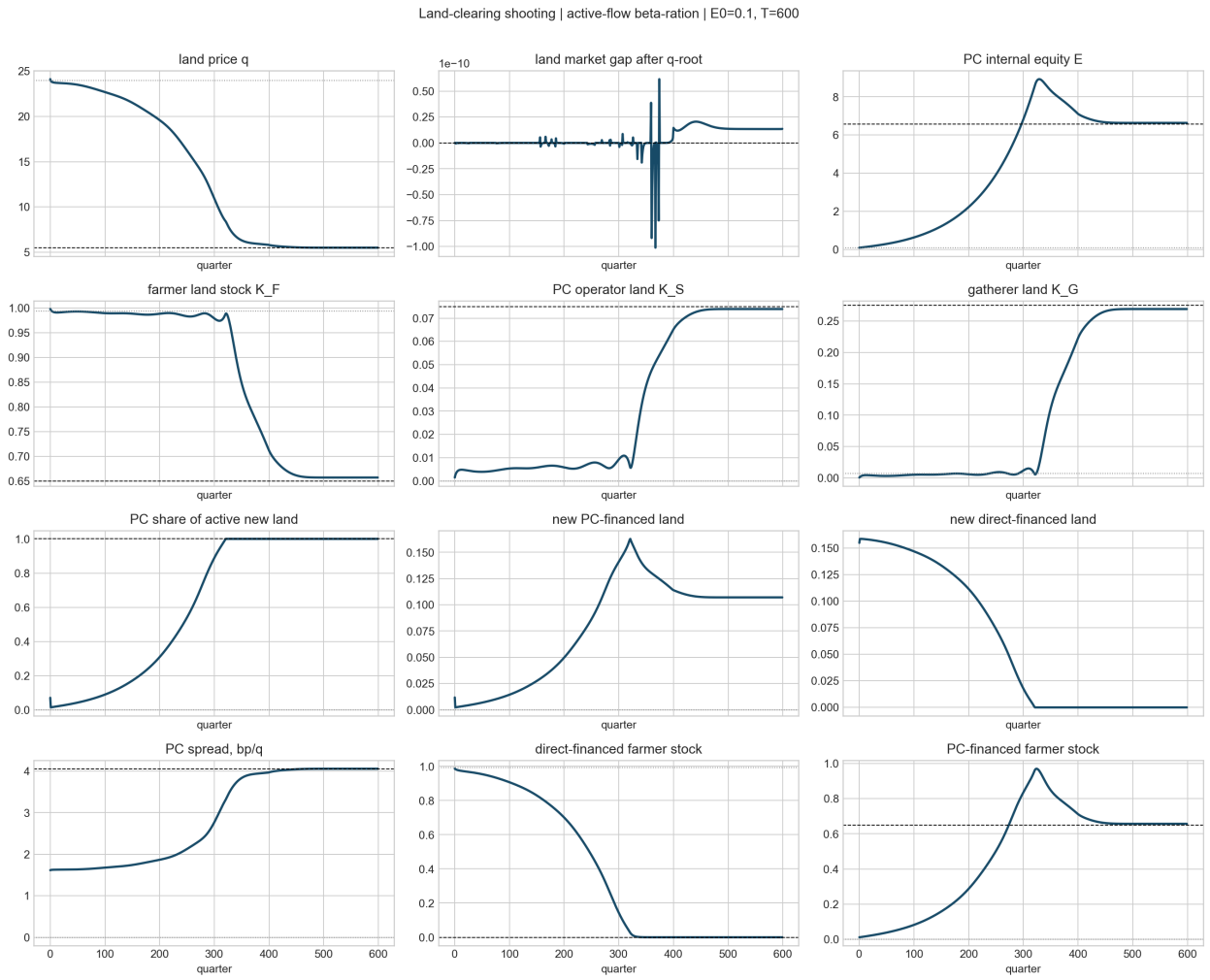


Figure 6: Transition following gardener entry

Notes: Perfect-foresight transition from the steady state without gardener lending to the gardener-lending steady state. Gardeners enter with initial internal equity $E_0 = 0.1$. The figure reports the land price, gardener equity, land holdings, the share and volume of new investment financed by gardeners, and the stocks of farmers financed by each source. Dashed horizontal lines indicate the terminal steady-state values where shown.

B Additional empirical results

B.1 Lender exposure

Figure 7 plots the distribution of E_ℓ, defined in equation (92), across the 72 BDCs with positive debt portfolios throughout 2019. Because C_k in equation (91) is negative for industries whose gross output rose in 2020, a BDC tilted toward those industries has negative exposure. The median exposure is -0.43 percentage points and the mean -0.25; the standard deviation is 3.90 and the 10th-to-90th percentile range runs from -4.47 to 3.24. Two BDCs lie far in the right tail, at roughly 10 and 19 percentage points. The ten-percentage-point exposure difference used to scale the coefficients in Table 2 equals 2.57 standard deviations of E_ℓ.

B.2 Industry shock and recovery

Figure 8 plots, for each of the 34 PitchBook IndustryGroups for which both measures are available, the COVID shock against the standardized recovery measure $\widetilde{R}_b$. The shock for a group is the 2019Q4 fair-value-weighted average of C_k across the IndustryCodes it contains. The correlation is 0.51. The industries hit hardest, energy and transportation, are those with the most transferable collateral; software, healthcare technology, and pharmaceuticals combine small or negative output losses with low recovery. Restaurants, hotels and leisure is the one large-shock, low-recovery group.

The sign of this correlation rules out a mechanical reading of β_h. A BDC is exposed because of the industries it holds. If exposed BDCs simply cut the industries responsible for their losses, they would cut high-recovery borrowers by more, and the interaction would be negative. The positive β_h in Table 2 is therefore reallocation across the balance sheet,

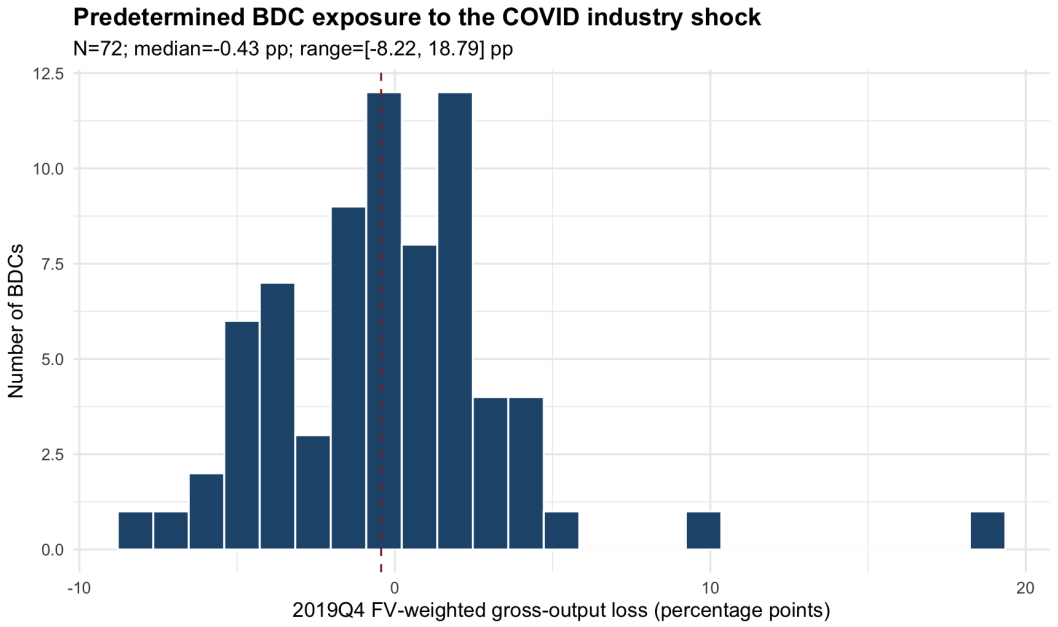


Figure 7: Distribution of predetermined BDC exposure. Histogram of E_ℓ across the 72 BDCs with positive debt portfolios in all four quarters of 2019. E_ℓ is the 2019Q4 fair-value-weighted average across BDC ℓ's debt positions of the 2019–2020 contraction in BEA gross output of the position's industry, in percentage points. The dashed line marks the median.

and own-industry deleveraging biases it toward zero. The leave-own-industry-out estimate in Table 3, 26.09 against 25.86 percentage points at 2021Q4, confirms that this bias is negligible.

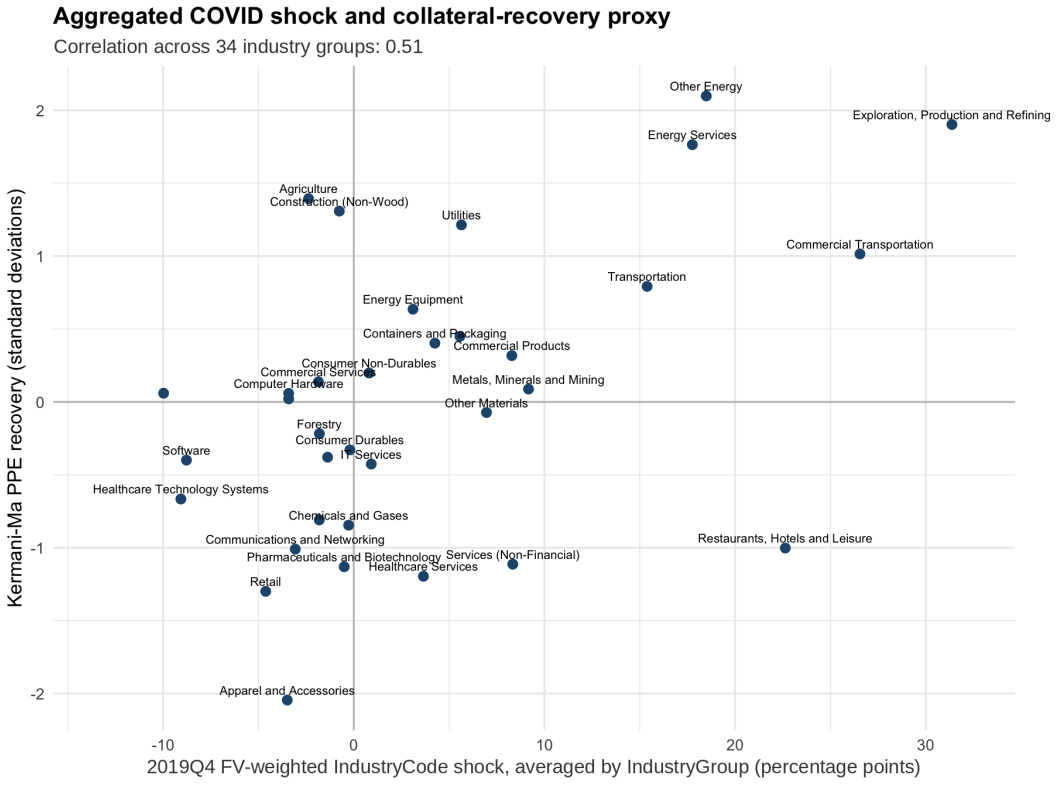


Figure 8: Industry-level COVID shock and collateral recovery. Each point is a PitchBook IndustryGroup. The horizontal axis is the 2019Q4 fair-value-weighted average across the group's IndustryCodes of the 2019–2020 contraction in BEA gross output, in percentage points. The vertical axis is the standardized Kermani and Ma (2023) property, plant, and equipment recovery rate $\widetilde{R}_b$. The correlation across the 34 groups is 0.51.

B.3 Exposure and sample robustness

Table 3 reports the 2021Q4 robustness estimates discussed in Section 4. The leave-own-industry-out measure removes the focal borrower's IndustryGroup from both the numerator and denominator of the BDC exposure measure. Principal-weighted exposure replaces

the 2019Q4 fair-value weights with principal weights. Repayment-adjusted principal adds back contractual principal reaching final maturity during the outcome window. The same-industry-and-location specification compares lenders across borrowers in the same PitchBook IndustryGroup and state or province, rather than within the same borrower; it also admits single-lender borrowers, so its difference from the baseline reflects both the fixed effects and the sample. The final two rows use equal regression weights and retain only continuing relationships.

Table 3: Cumulative credit growth to 2021Q4: robustness

	$\widehat{\beta}_7$	S.E.	BDC-WCB p	Observations
Baseline exposure	25.86	9.33	0.007	4,266
Leave-own-IndustryGroup-out exposure	26.09	11.64	0.017	4,266
Principal-weighted exposure	23.57	8.34	0.010	4,266
Repayment-adjusted principal	20.37	10.58	0.074	4,266
Same-industry-and-location comparison	10.26	10.67	0.368	6,768
Equal regression weights	15.27	9.67	0.187	4,266
Continuing relationships only	4.69	6.74	0.508	1,250

Notes: The table reports the exposure-by-recovery coefficient at $h = 7$. All regressions include the lower-order exposure term. Except in the equal-weight row, observations receive baseline-endpoint midpoint portfolio-share weights. Standard errors are two-way clustered by BDC and borrower. BDC-WCB p-values are two-sided wild-cluster-bootstrap tests with 9,999 Rademacher replications, BDC bootstrap weights, and BDC–borrower two-way studentization. Leave-own-IndustryGroup-out exposure removes the focal borrower's IndustryGroup from both the numerator and denominator of the BDC exposure measure. Principal-weighted exposure replaces fair-value weights with par-value principal weights. Repayment-adjusted principal adds contractual principal reaching final maturity during the outcome window; it does not adjust for interim amortization, voluntary prepayment, or refinancing. The same-industry-and-location row replaces borrower fixed effects with PitchBook IndustryGroup-by-state-or-province fixed effects and admits single-lender borrowers. All other rows include borrower fixed effects. The final row retains only relationships with positive principal in both 2019Q4 and 2021Q4.

B.4 Earlier-baseline placebo

As a further check on pre-trends, we apply the same eight-quarter long-difference design to 2017Q4–2019Q4, measuring each BDC's exposure with its 2017Q4 portfolio shares in the

industries subsequently hit by COVID-19. Table 4 reports the estimate. The interaction is −15.94 percentage points, with a BDC-margin wild-bootstrap p-value of 0.098. It is opposite in sign to the post-COVID estimate: over 2017Q4–2019Q4, BDCs whose portfolios were tilted toward the industries that COVID-19 would later hit grew less toward high-recovery borrowers, so a continuation of pre-COVID lending patterns would produce a negative interaction after 2019Q4, not the positive one we find. The set of multi-lender borrowers is defined over the full 2018Q1–2024Q1 sample, as in the main analysis; the placebo's exposure and outcome use no information after 2019Q4.

Table 4: Horizon-matched pre-COVID placebo

	2017Q4–2019Q4
Exposure × recovery	−15.94*
BDC–borrower clustered S.E.	(7.95)
Conventional p-value	[0.049]
BDC-WCB p-value	[0.098]
Observations	4,498
BDCs	63
Borrowers	1,484
Borrower fixed effects	Yes
Midpoint portfolio-share weights	Yes

Notes: The outcome is symmetric principal growth from 2017Q4 to 2019Q4, matching the eight-quarter length of the main 2019Q4–2021Q4 comparison. Exposure uses 2017Q4 fair-value portfolio shares applied to the eventual 2019–2020 industry output shocks. Recovery is the standardized IndustryGroup measure. The BDC-WCB test uses BDC–borrower studentization and 9,999 BDC-level Rademacher draws. The star uses the bootstrap p-value: $^{*}p < 0.10$.

B.5 Alternative recovery measures

We consider two alternatives to the IndustryGroup recovery measure of the main specification. The first assigns each borrower a canonical PitchBook IndustryCode and averages directly matched borrower recovery within that code. A code uses its own mean when at

least three borrowers have matched recovery and otherwise receives the mean of its parent IndustryGroup. This produces 174 IndustryCode cells, of which 137 use their own mean and 37 the parent-group mean. Recovery is then standardized with equal weight across the 174 cells.

The second alternative assigns each borrower the recovery of its matched two-digit SIC industries, averaged without weights when there are several. This measure is standardized across the 3,681 borrowers with a direct match and leaves unmatched borrowers out. Table 5 reports the corresponding within-borrower estimates, and Figures 9 and 10 the complete coefficient paths. The IndustryCode path is indistinguishable from the baseline. The direct SIC measure, on a sample less than half the size, gives a larger and less precise interaction at 2021Q4, 35.79 percentage points with a bootstrap p-value of 0.060, and a larger exposure main effect.

Table 5: Within-borrower estimates using alternative recovery measures

	2020Q4 $h = 3$	2021Q4 $h = 7$	2024Q1 $h = 16$
Panel A: IndustryCode recovery			
Exposure × recovery, β_h	5.35	25.28***	6.26
	(7.60)	(9.43)	(12.75)
BDC-WCB p-value	[0.497]	[0.010]	[0.700]
Exposure, γ_h	−30.19***	−41.93***	−11.08
	(11.70)	(13.20)	(23.77)
BDC-WCB p-value	[0.002]	[0.004]	[0.793]
Observations	4,022	4,266	4,556
Borrowers	1,387	1,501	1,566
Panel B: Direct SIC recovery			
Exposure × recovery, β_h	−8.94	35.79*	21.68
	(13.81)	(16.30)	(17.48)
BDC-WCB p-value	[0.534]	[0.060]	[0.209]
Exposure, γ_h	−38.27***	−75.66***	−9.02
	(15.16)	(18.93)	(29.13)
BDC-WCB p-value	[0.002]	[< 0.001]	[0.865]
Observations	2,126	2,010	2,041
Borrowers	643	632	638
Borrower fixed effects	Yes	Yes	Yes
Midpoint portfolio-share weights	Yes	Yes	Yes

Notes: Each column is a separate endpoint cross-section. Panel A aggregates and standardizes recovery across PitchBook IndustryCodes. Panel B assigns recovery directly from matched two-digit SIC industries and standardizes it across matched borrowers. All other definitions and inference follow Table 2.

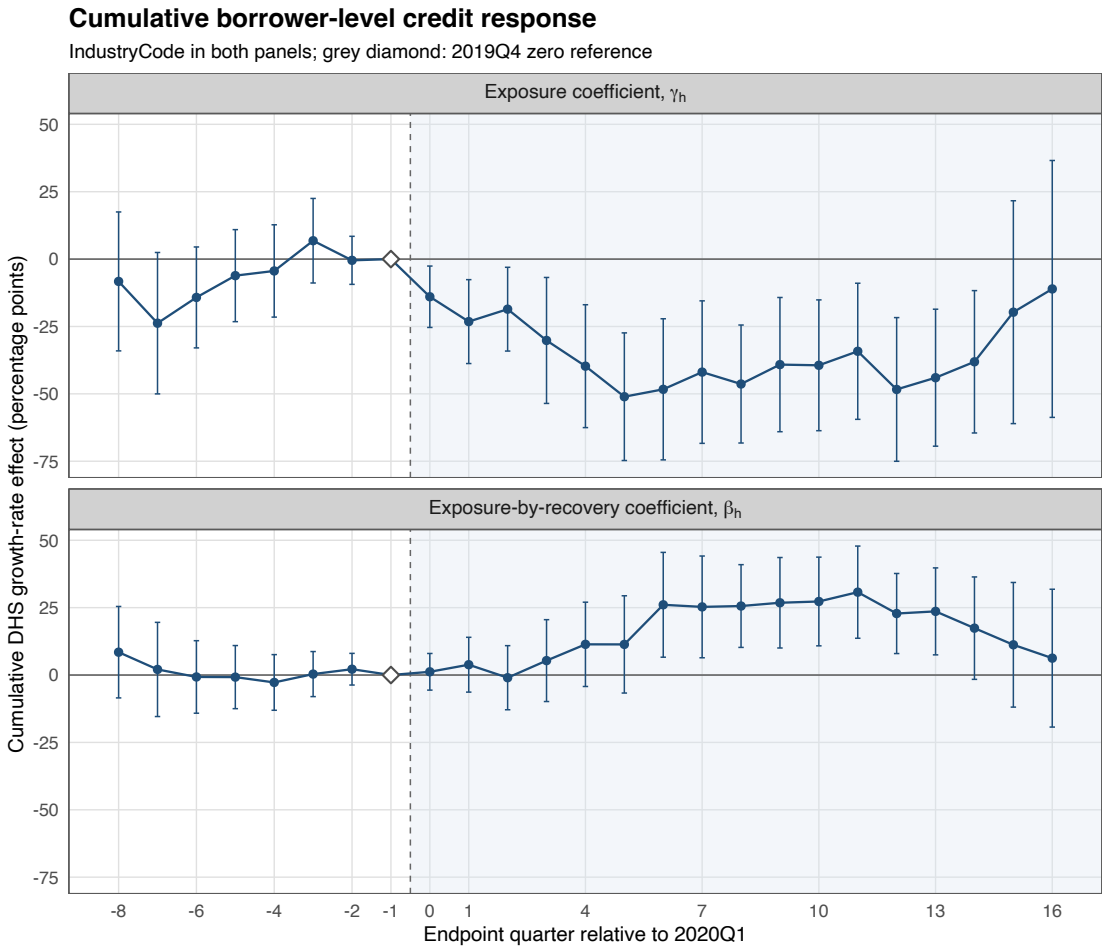


Figure 9: Cumulative within-borrower coefficients using IndustryCode recovery. The top panel plots the exposure coefficient $\widehat{\gamma}_h$ and the bottom panel plots the exposure-by-recovery coefficient $\widehat{\beta}_h$. The specification uses midpoint portfolio-share weights and 95 percent confidence intervals based on standard errors two-way clustered by BDC and borrower.

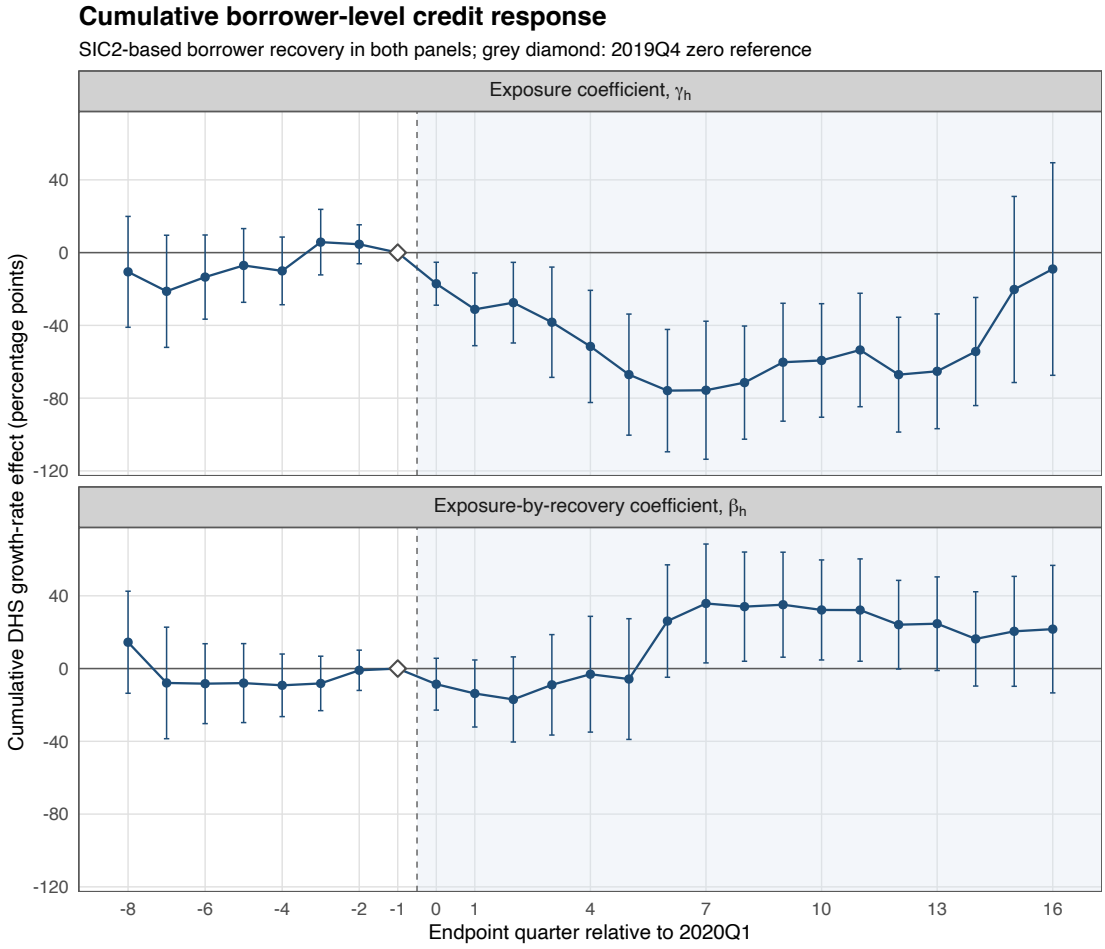


Figure 10: Cumulative within-borrower coefficients using direct SIC-based borrower recovery. The top panel plots the exposure coefficient $\widehat{\gamma}_h$ and the bottom panel plots the exposure-by-recovery coefficient $\widehat{\beta}_h$. The specification uses midpoint portfolio-share weights and 95 percent confidence intervals based on standard errors two-way clustered by BDC and borrower.